

BFS Traversal

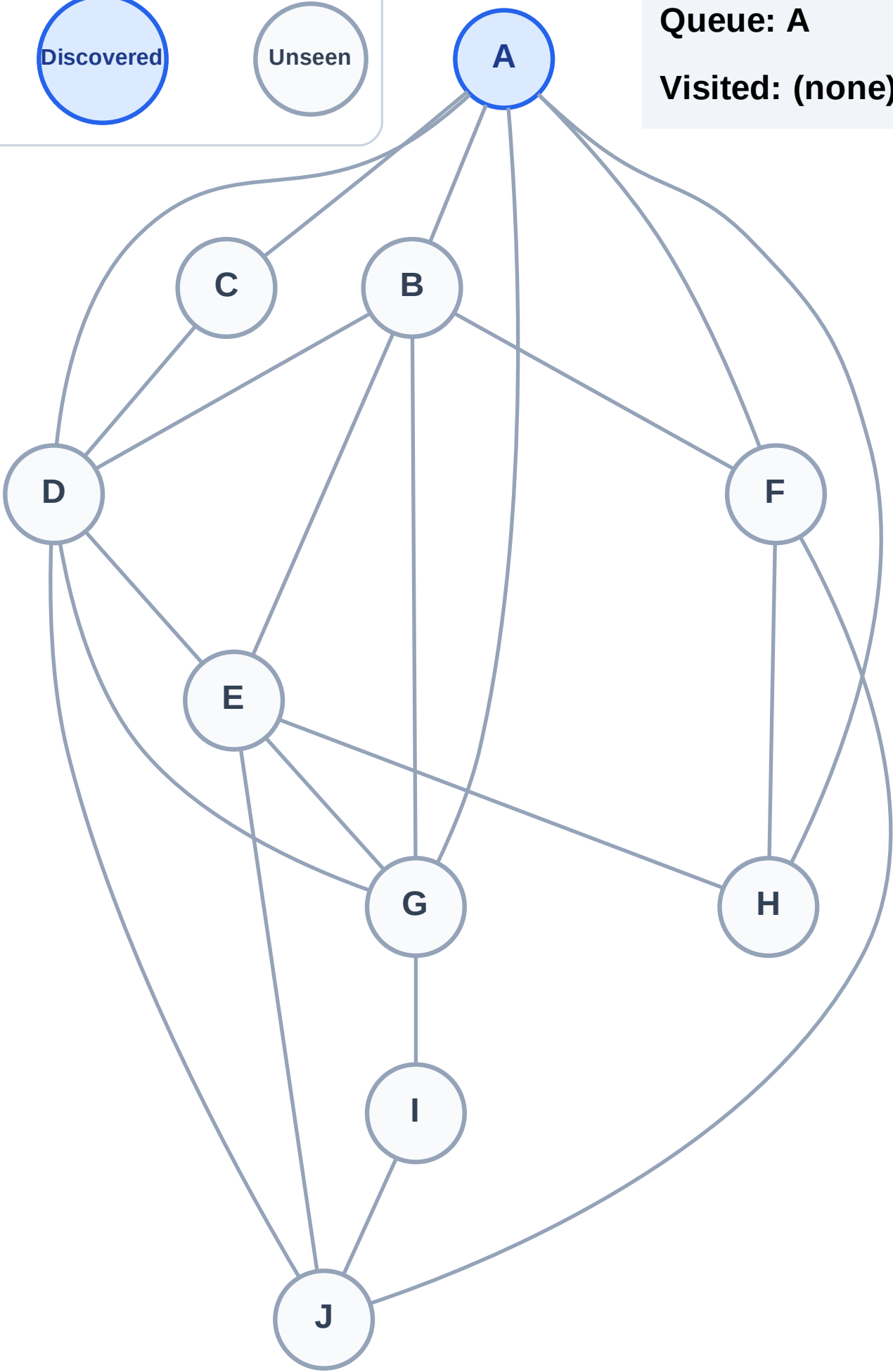
Step 1: Start at A

Legend



Queue: A

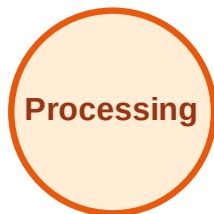
Visited: (none)



BFS Traversal

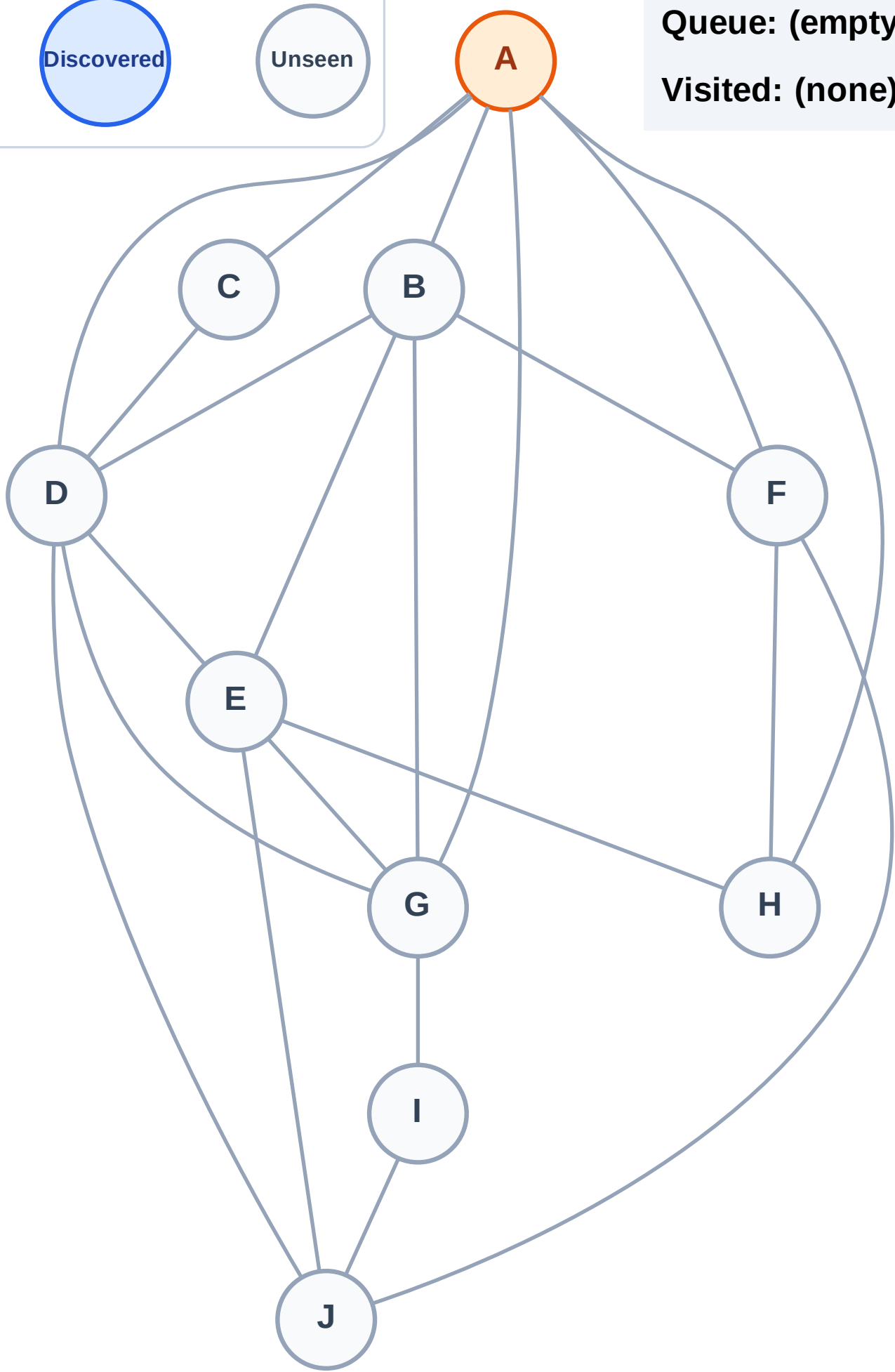
Step 2: Process node A

Legend



Queue: (empty)

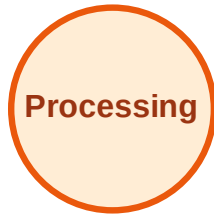
Visited: (none)



BFS Traversal

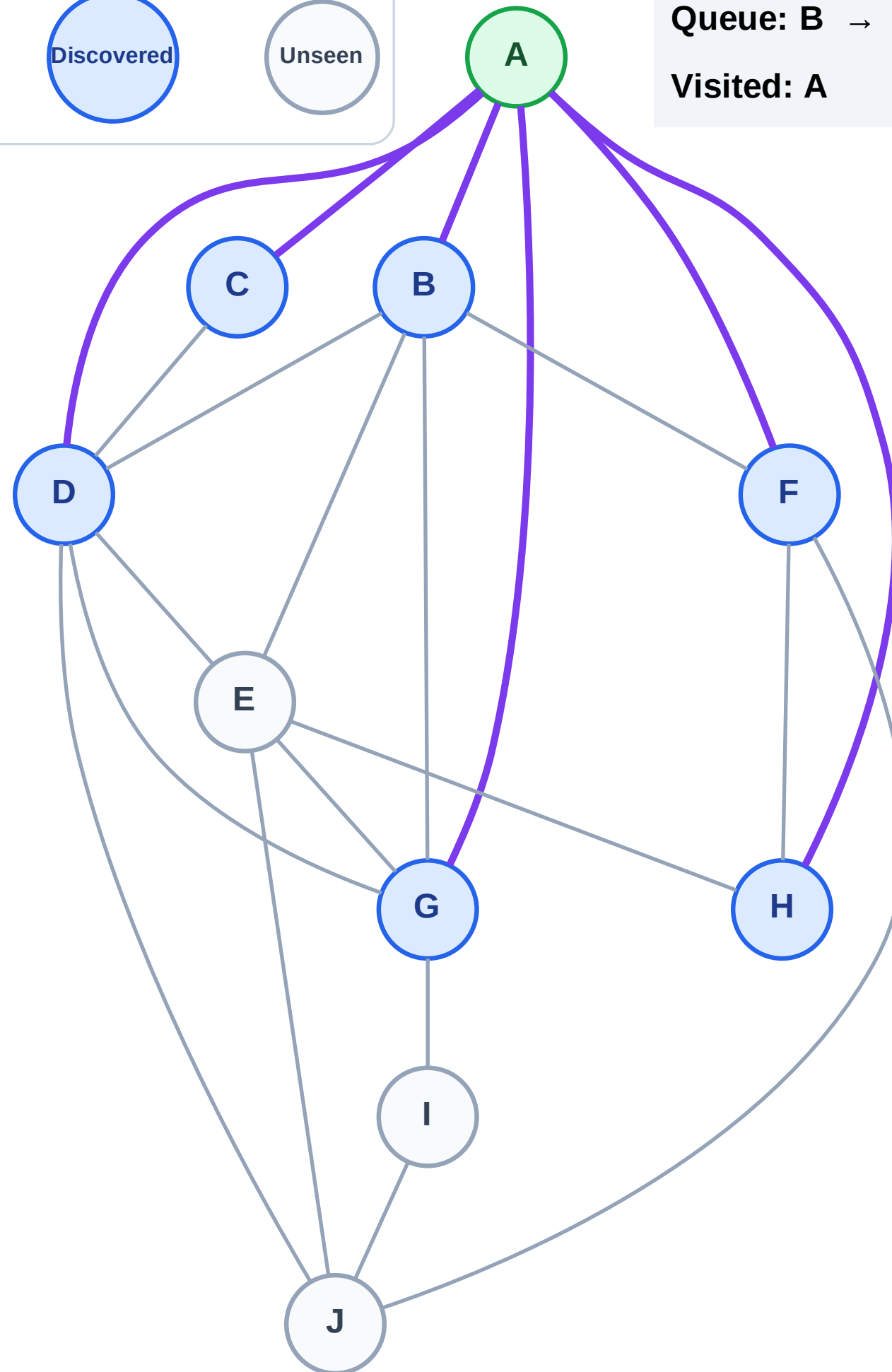
Step 3: Discover B, C, D, F, G, H from A

Legend



Queue: B → C → D → F → G → H

Visited: A



BFS Traversal

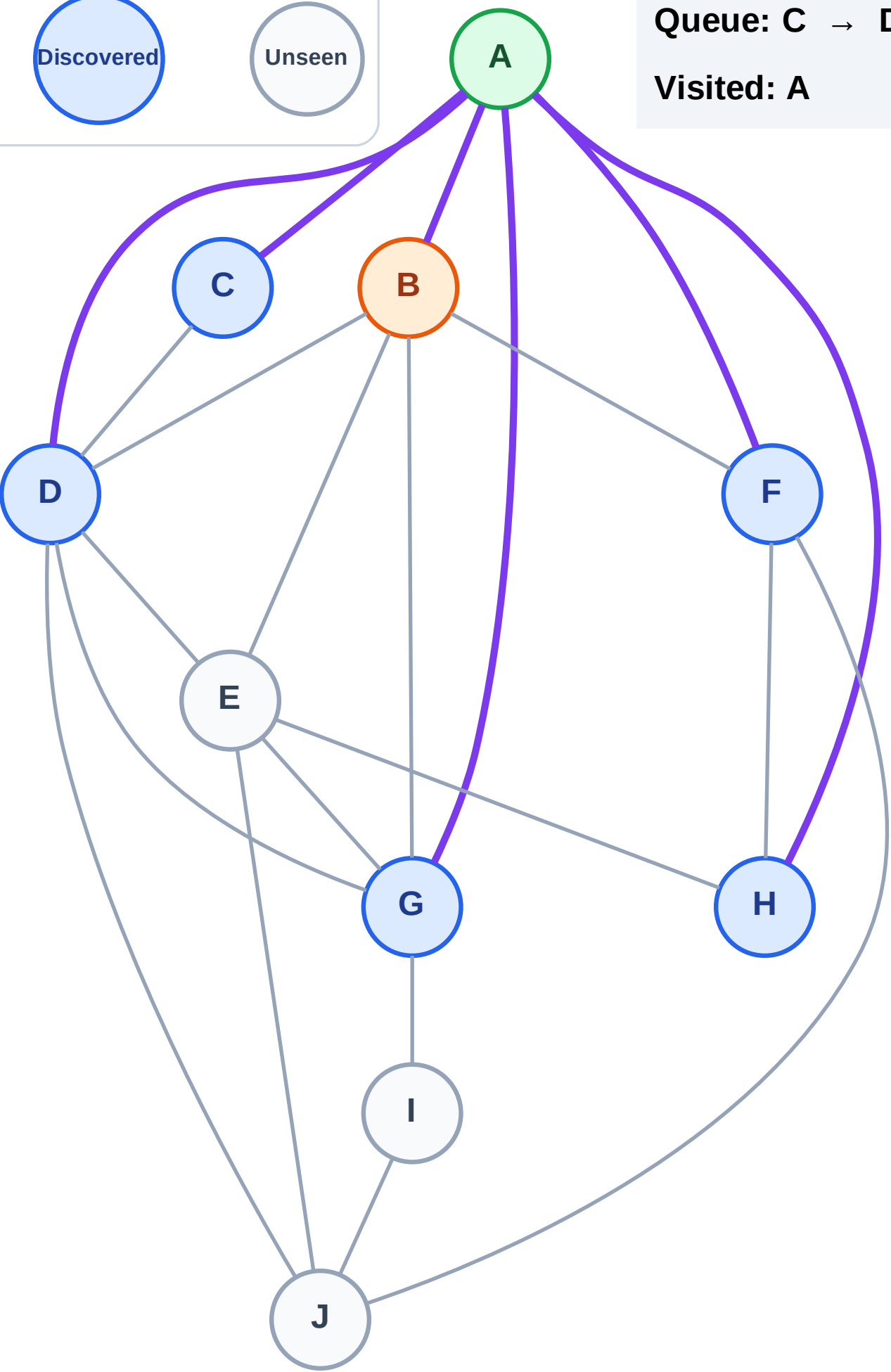
Step 4: Process node B

Legend



Queue: C → D → F → G → H

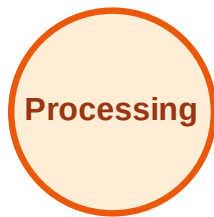
Visited: A



BFS Traversal

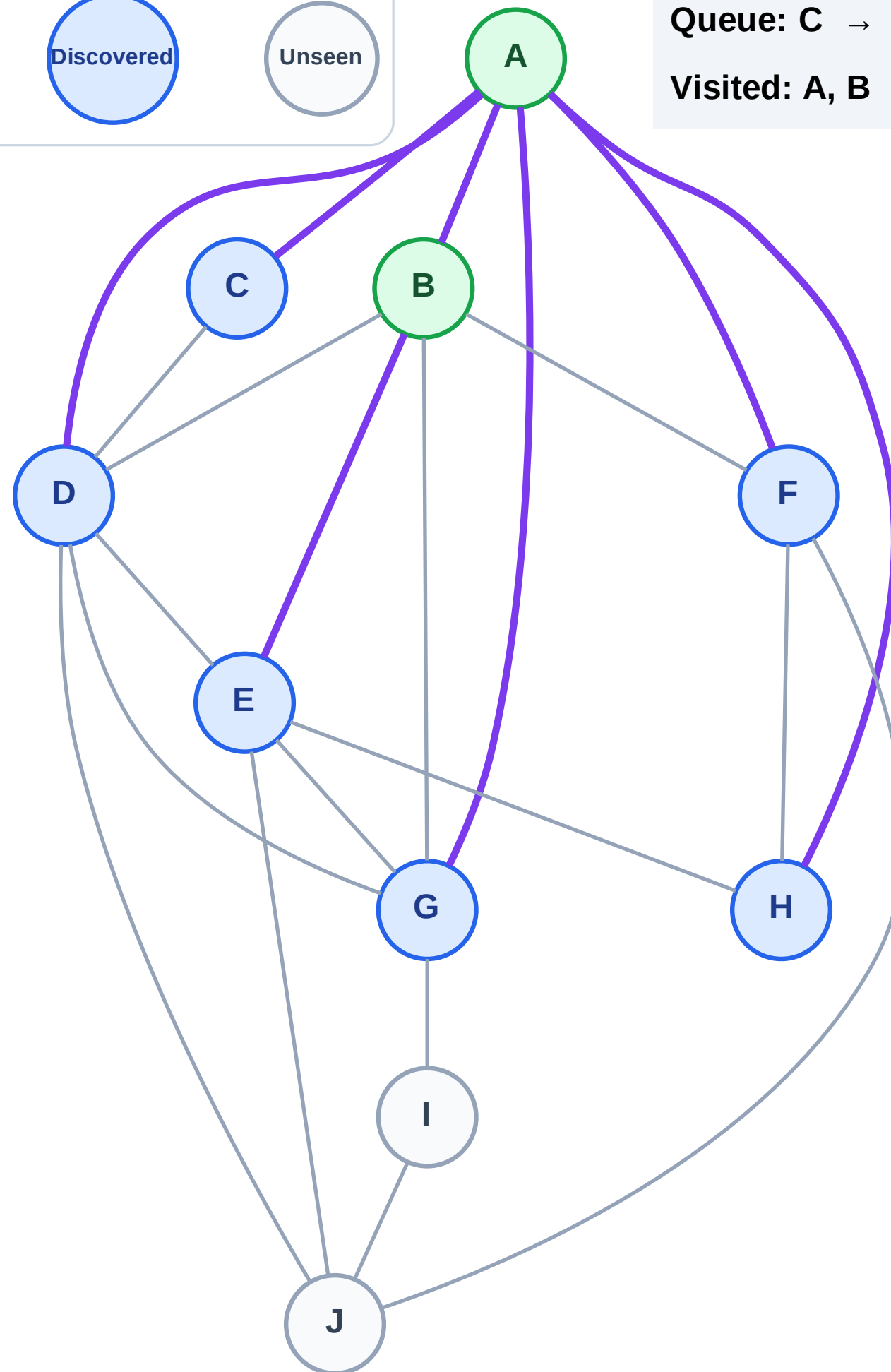
Step 5: Discover E from B

Legend



Queue: C → D → F → G → H → E

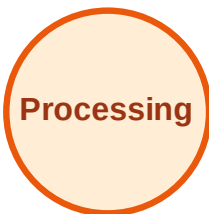
Visited: A, B



BFS Traversal

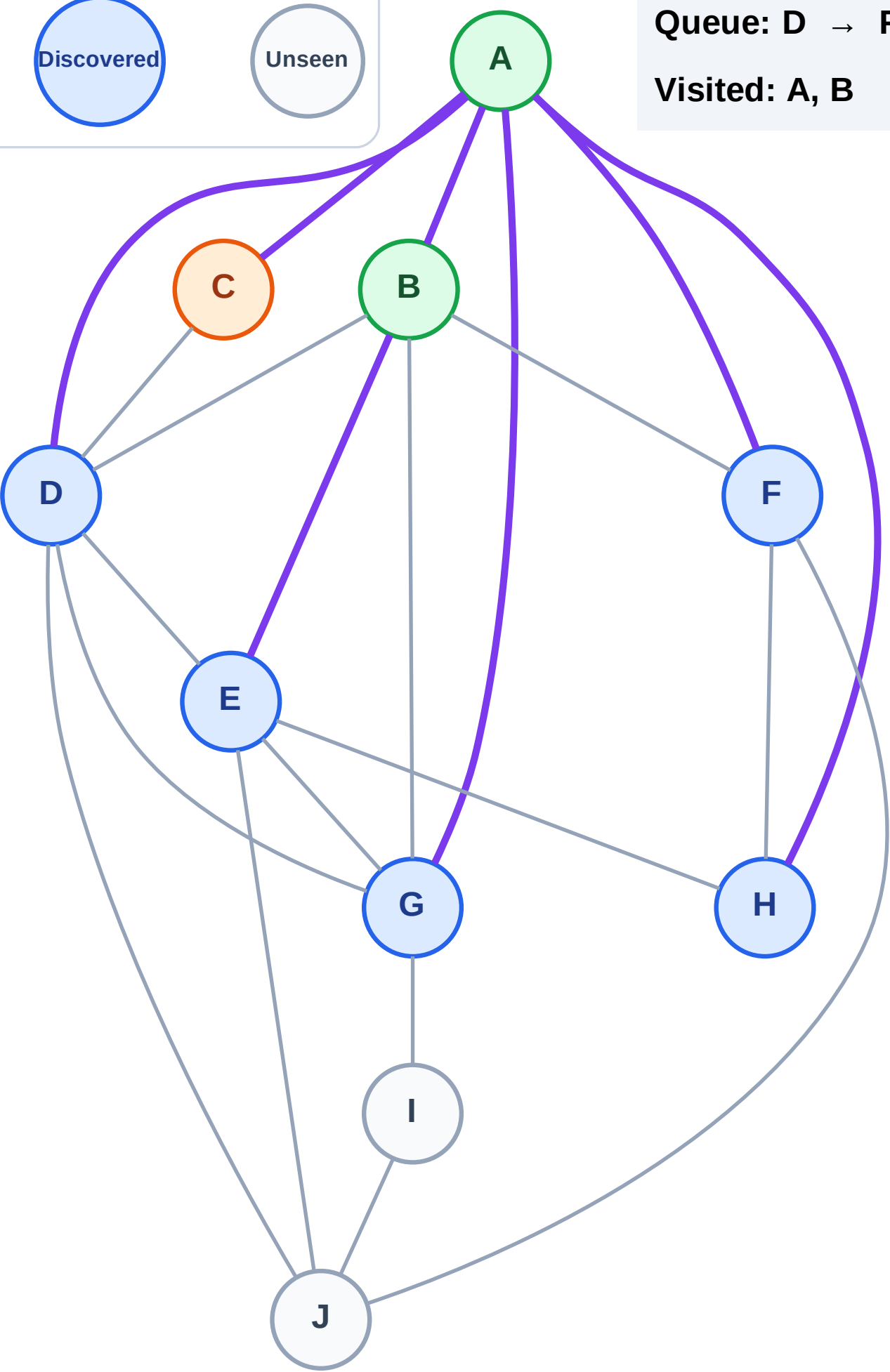
Step 6: Process node C

Legend



Queue: D → F → G → H → E

Visited: A, B



BFS Traversal

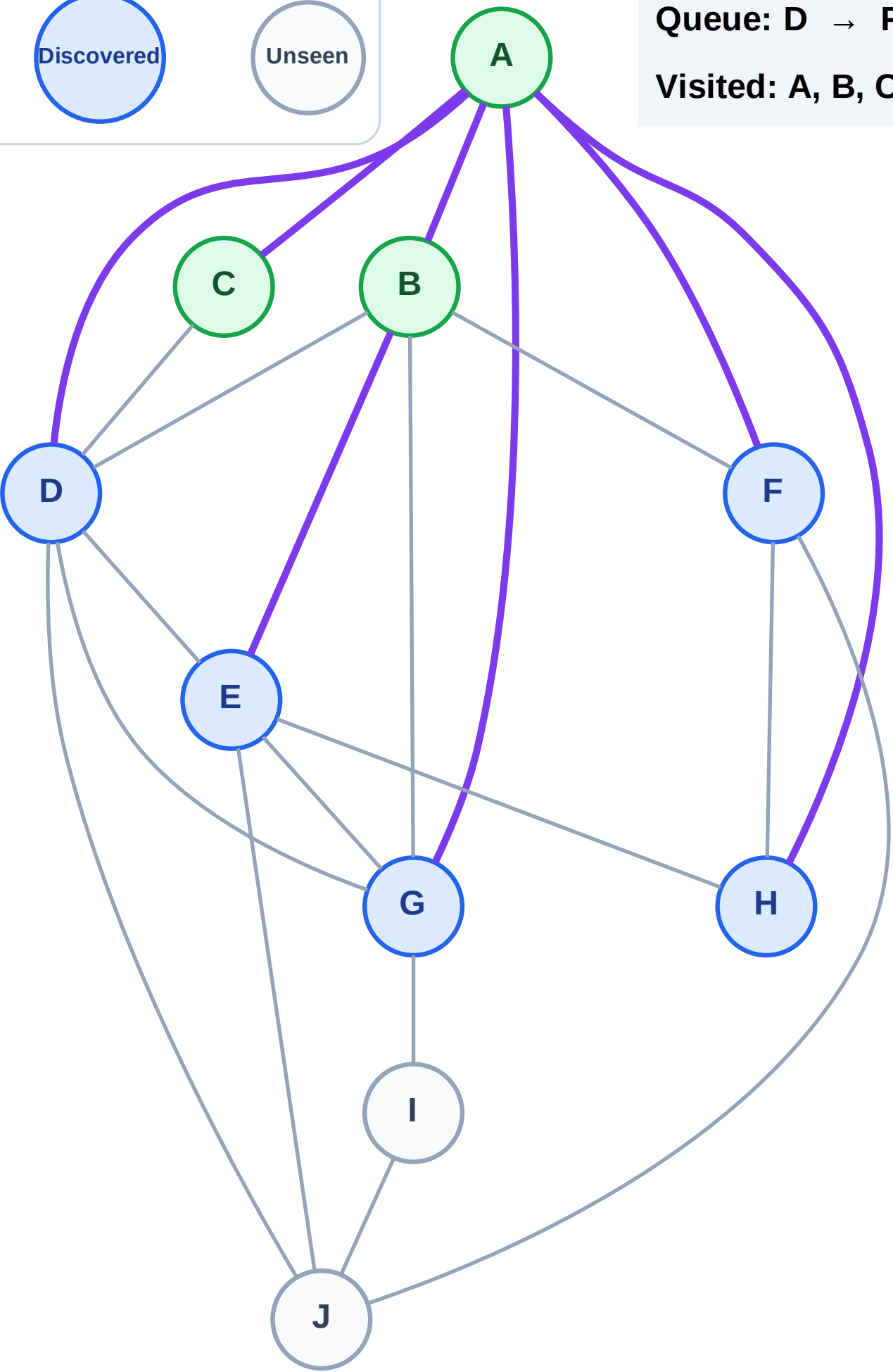
Step 7: No new neighbors from C

Legend



Queue: D → F → G → H → E

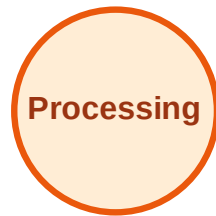
Visited: A, B, C



BFS Traversal

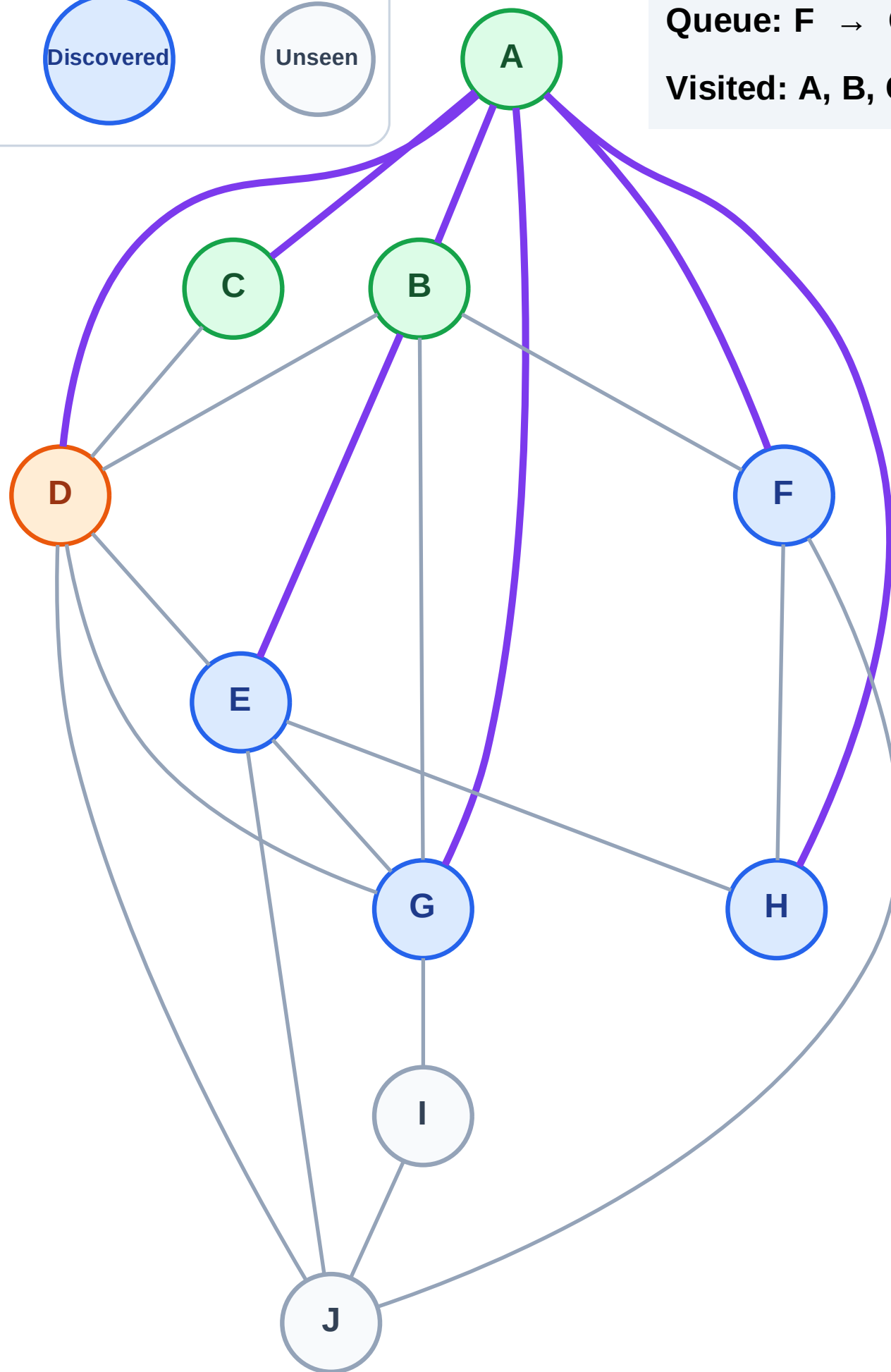
Step 8: Process node D

Legend



Queue: F → G → H → E

Visited: A, B, C



BFS Traversal

Step 9: Discover J from D

Legend



Complete



Processing



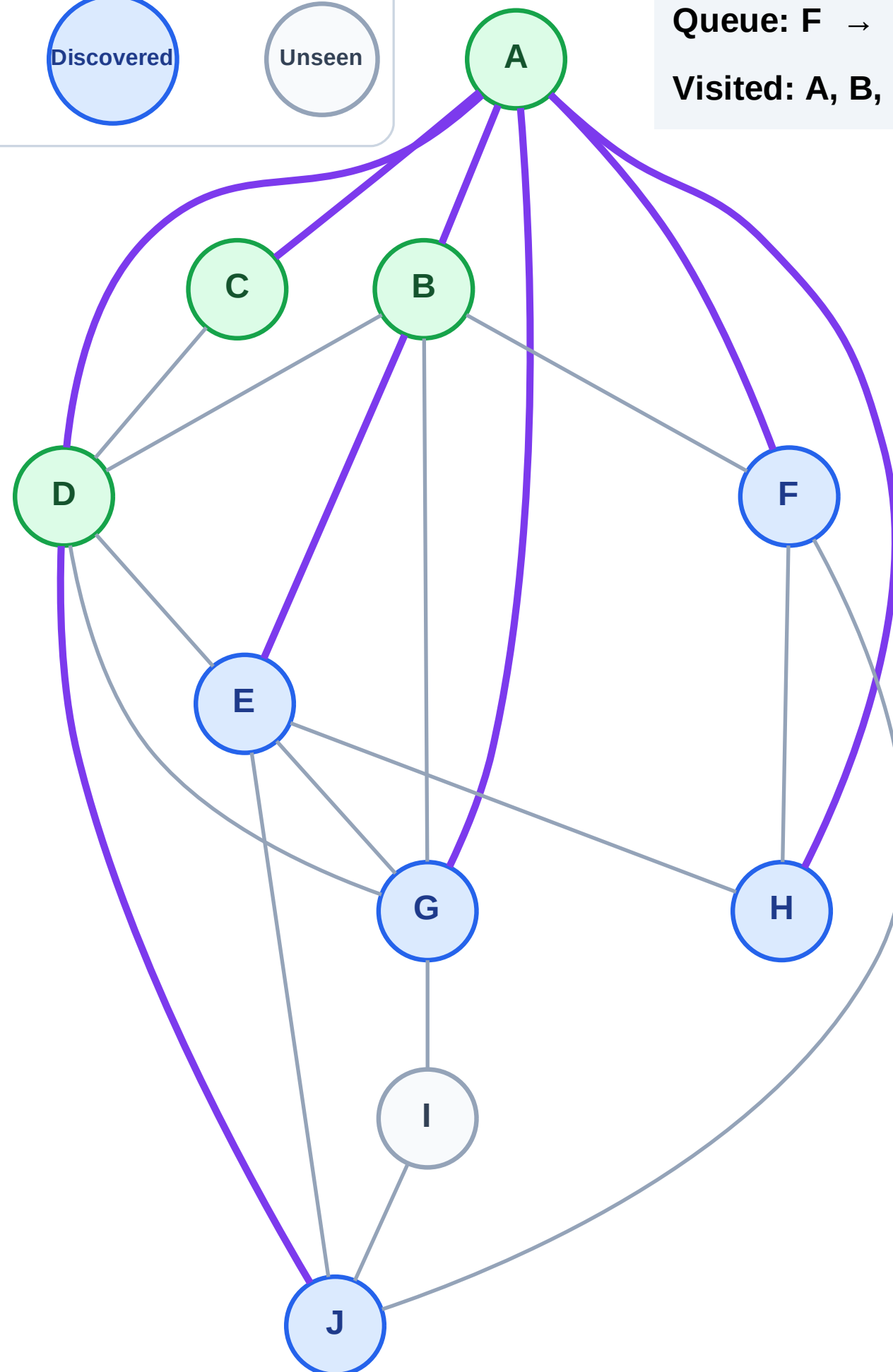
Discovered



Unseen

Queue: F → G → H → E → J

Visited: A, B, C, D



BFS Traversal

Step 10: Process node F

Legend

Complete

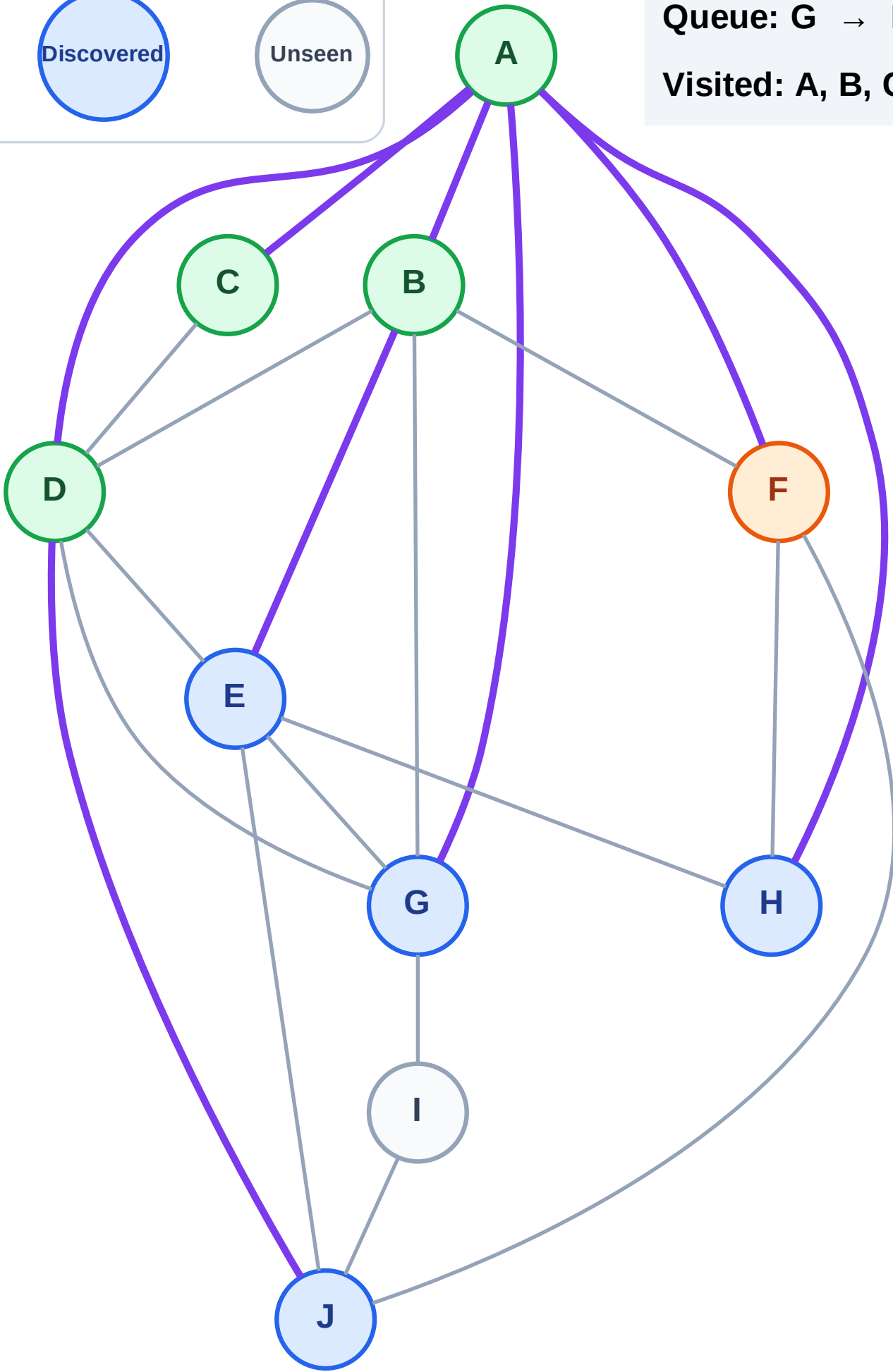
Processing

Discovered

Unseen

Queue: G → H → E → J

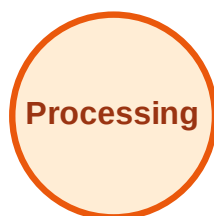
Visited: A, B, C, D



BFS Traversal

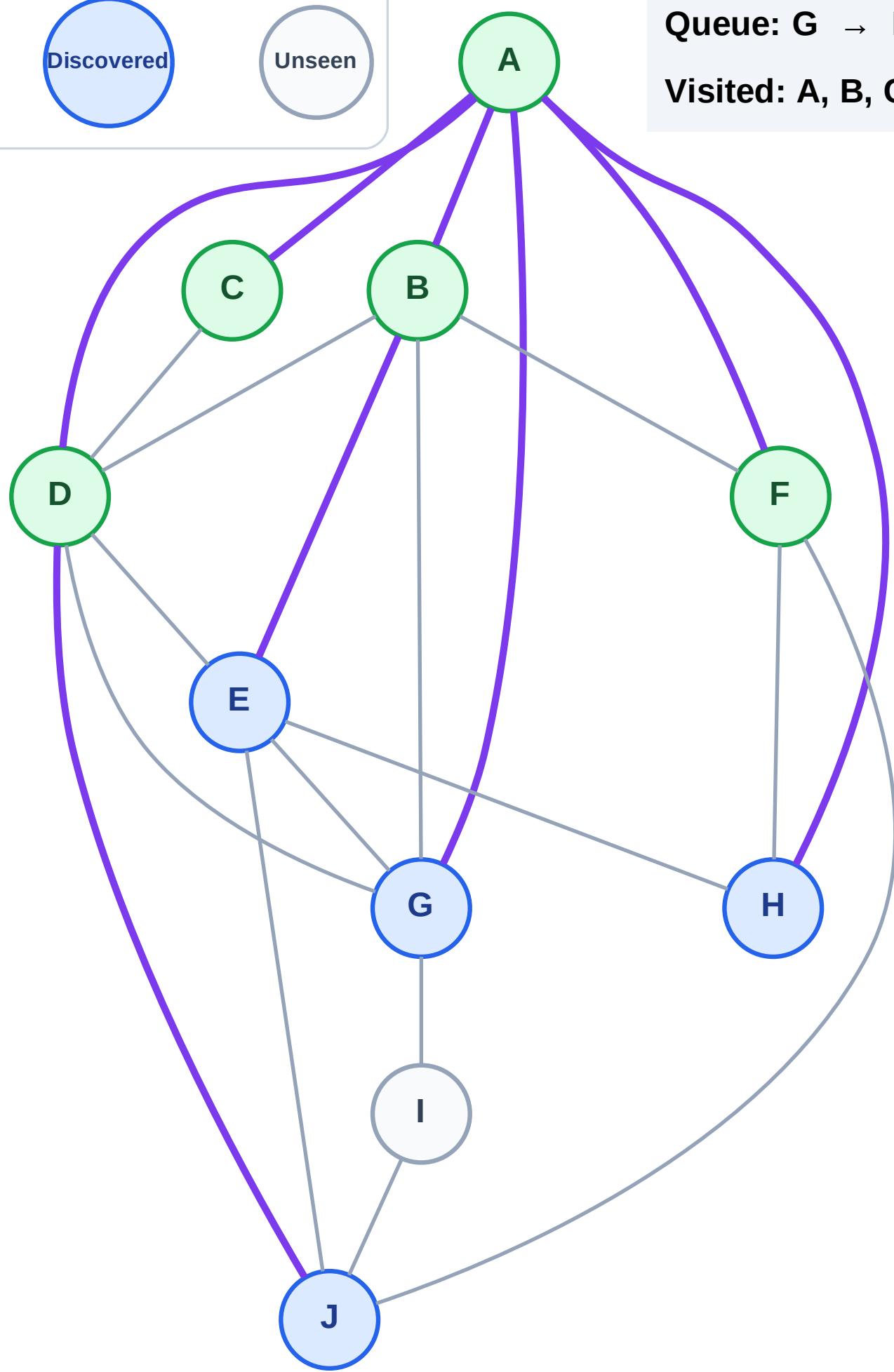
Step 11: No new neighbors from F

Legend



Queue: G → H → E → J

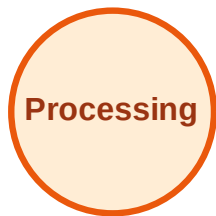
Visited: A, B, C, D, F



BFS Traversal

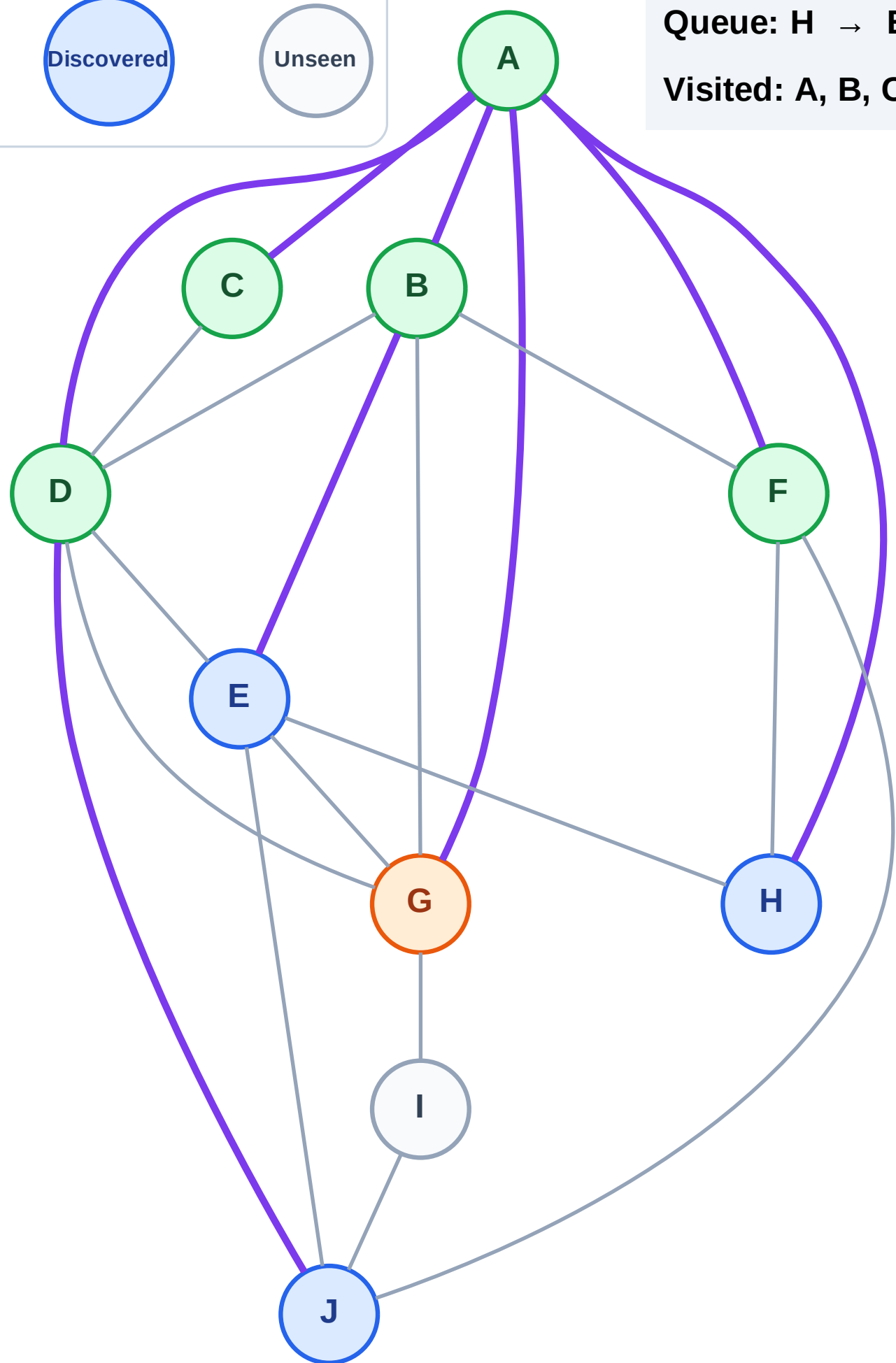
Step 12: Process node G

Legend



Queue: H → E → J

Visited: A, B, C, D, F



BFS Traversal

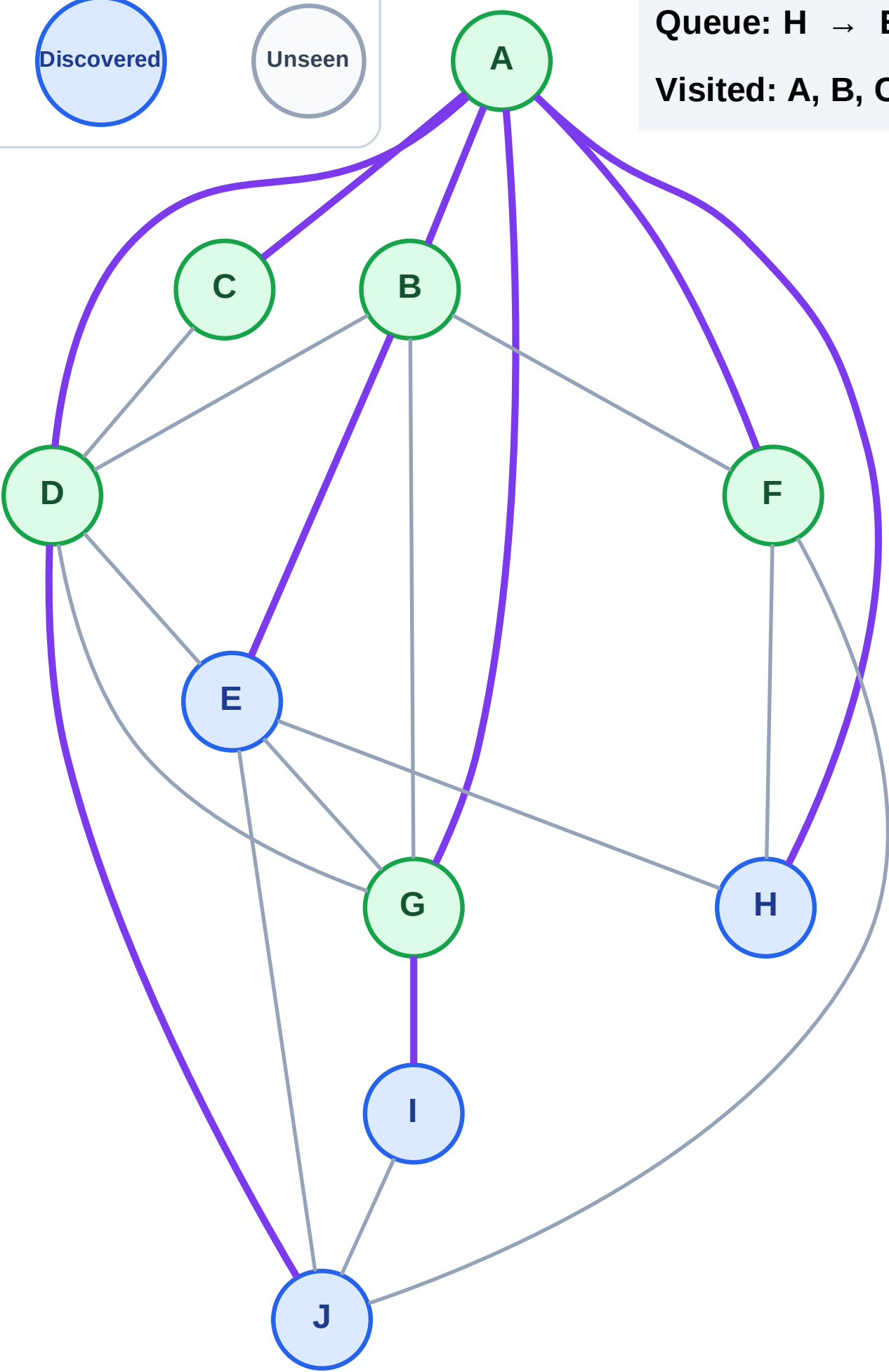
Step 13: Discover I from G

Legend



Queue: H → E → J → I

Visited: A, B, C, D, F, G



BFS Traversal

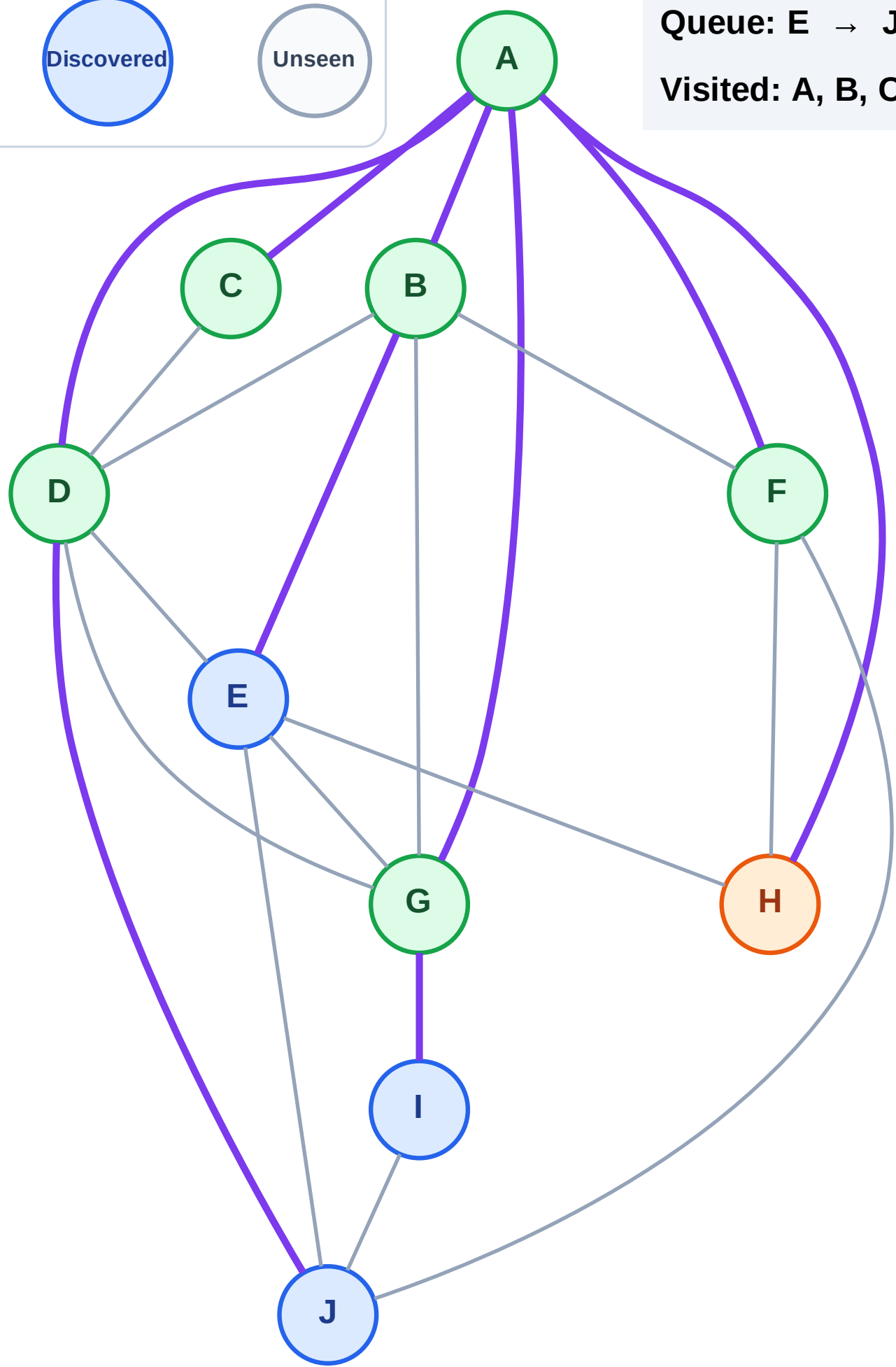
Step 14: Process node H

Legend



Queue: E → J → I

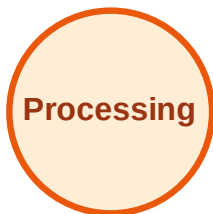
Visited: A, B, C, D, F, G



BFS Traversal

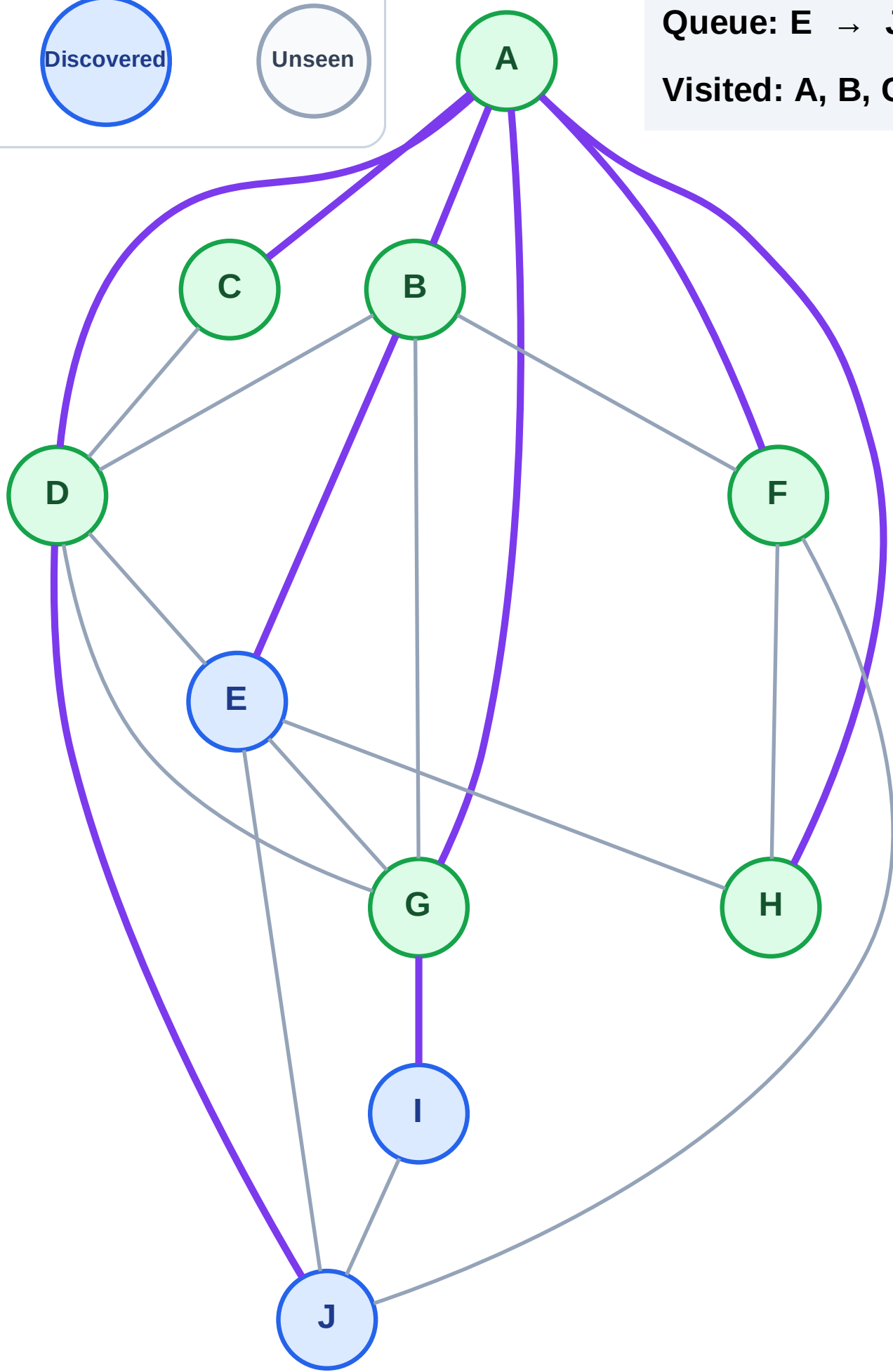
Step 15: No new neighbors from H

Legend

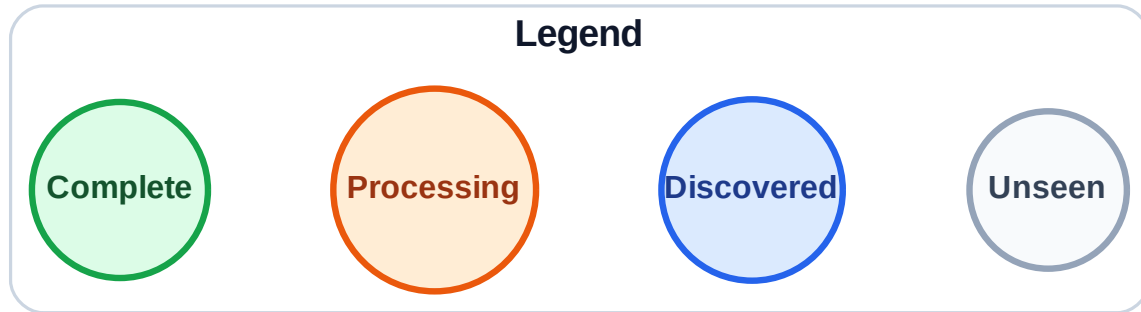


Queue: E → J → I

Visited: A, B, C, D, F, G, H

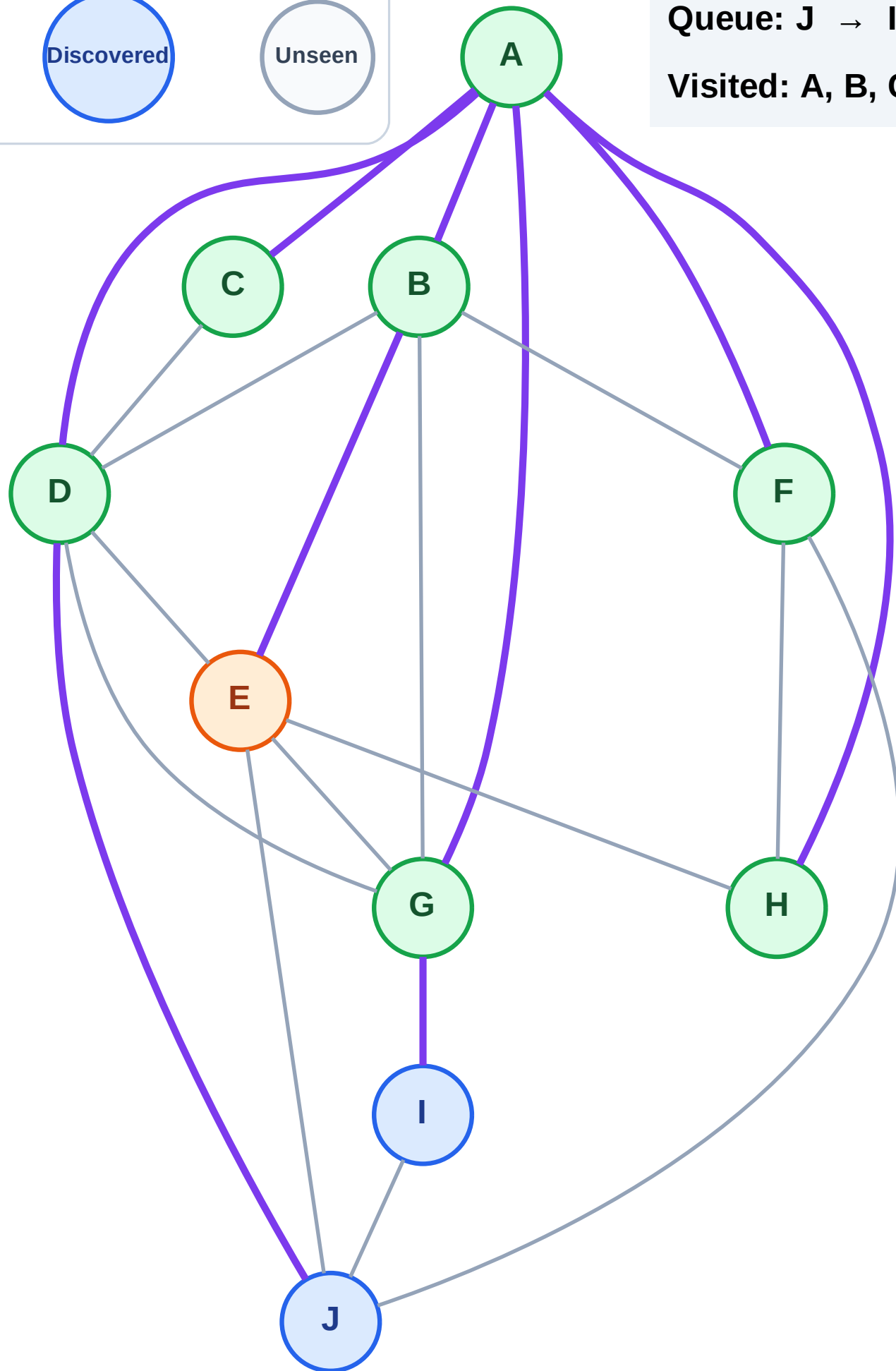


Step 16: Process node E



Queue: J → I

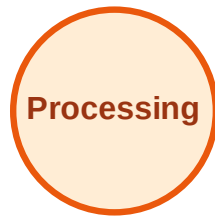
Visited: A, B, C, D, F, G, H



BFS Traversal

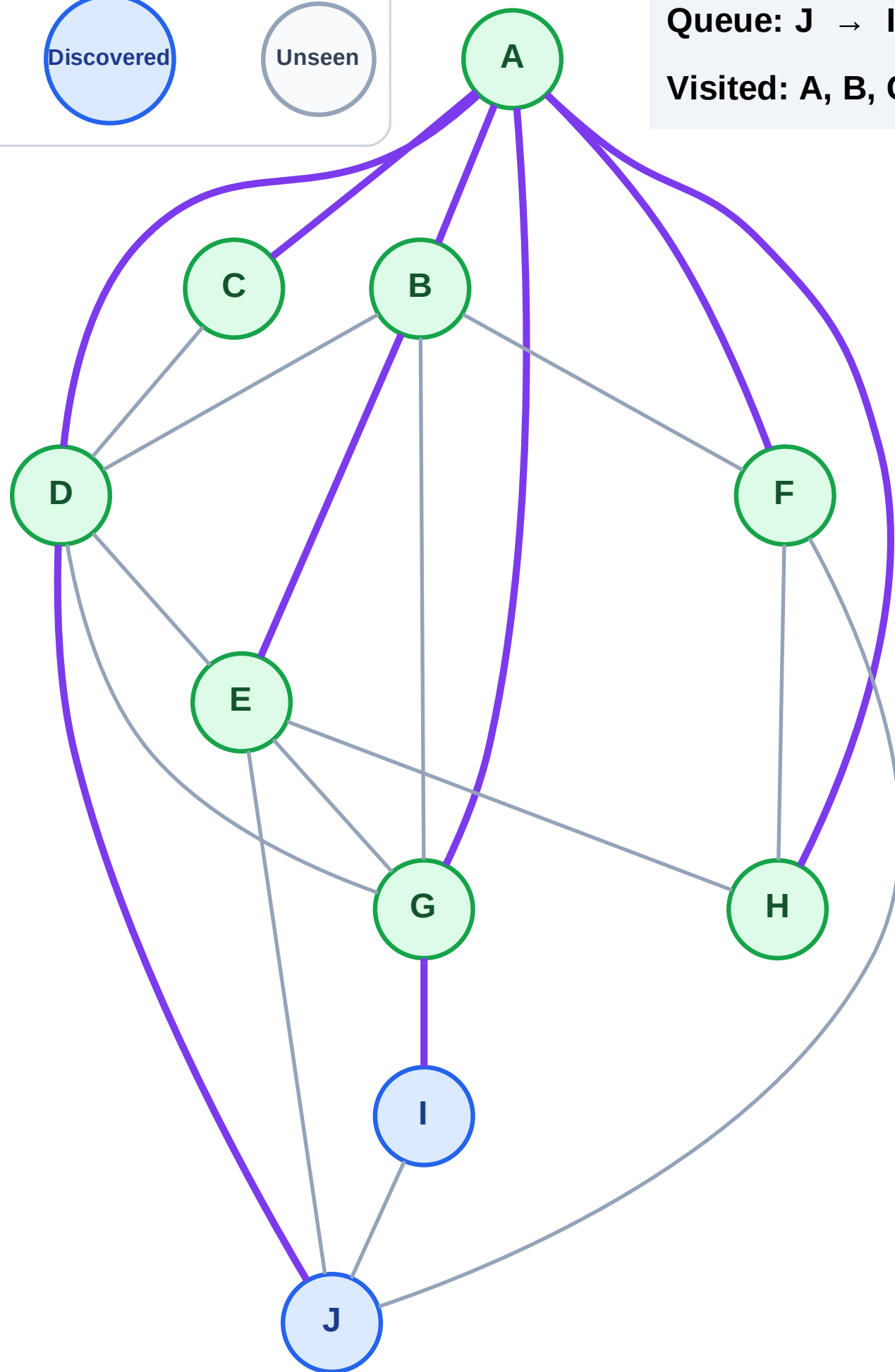
Step 17: No new neighbors from E

Legend



Queue: J → I

Visited: A, B, C, D, E, F, G, H



BFS Traversal

Step 18: Process node J

Legend

Complete

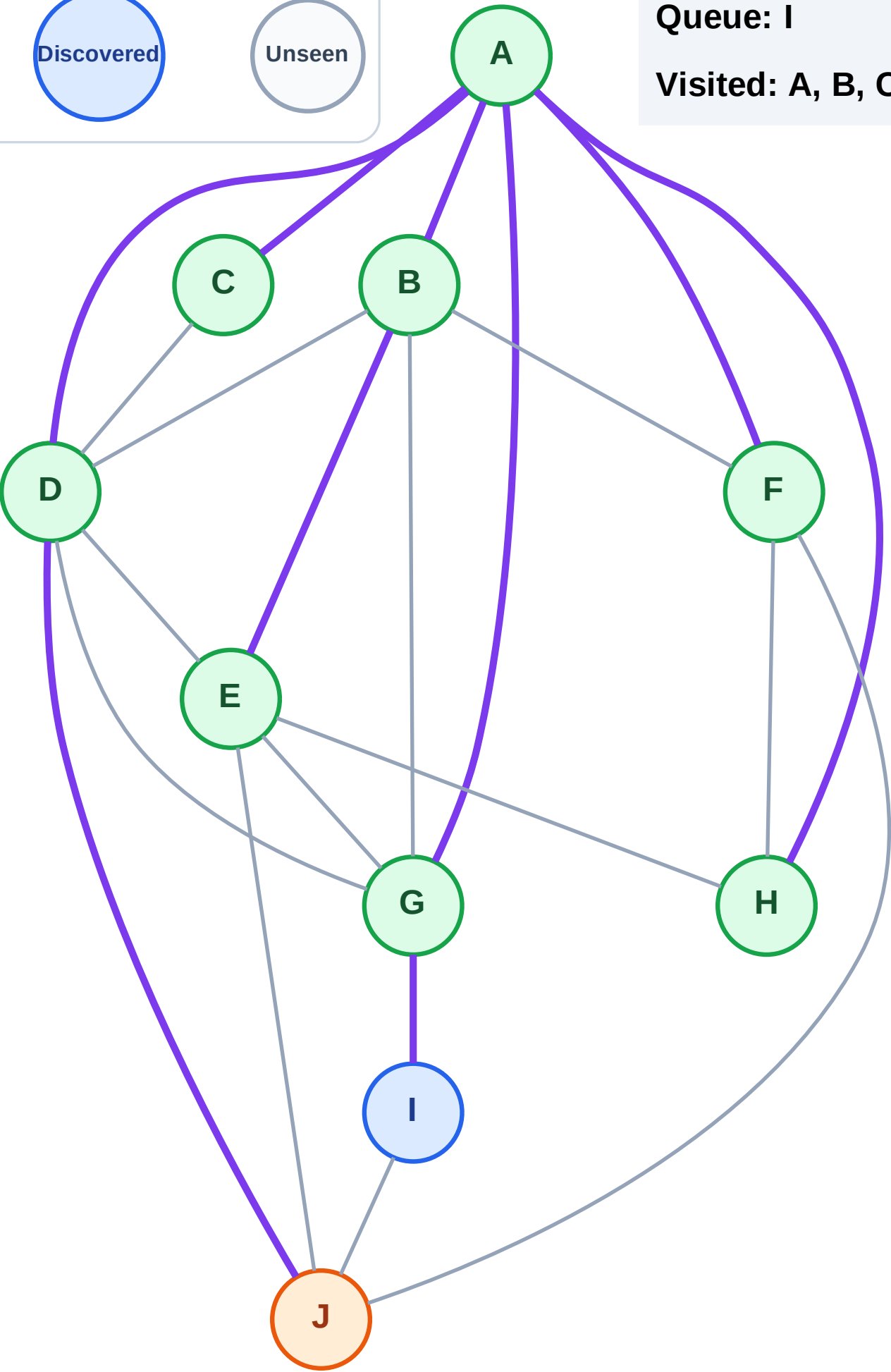
Processing

Discovered

Unseen

Queue: I

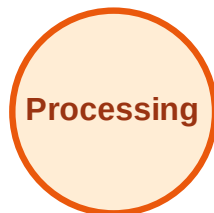
Visited: A, B, C, D, E, F, G, H



BFS Traversal

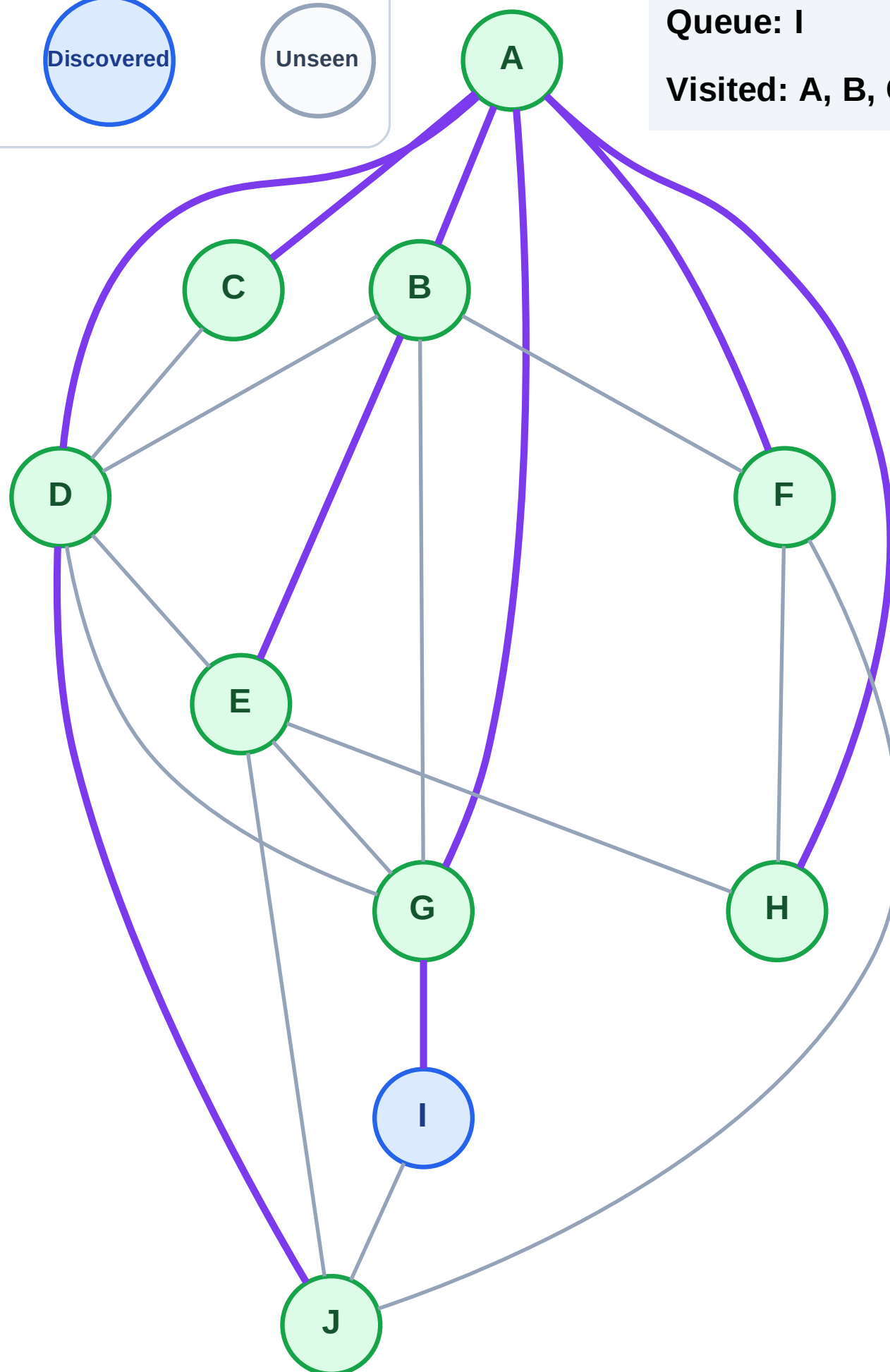
Step 19: No new neighbors from J

Legend



Queue: I

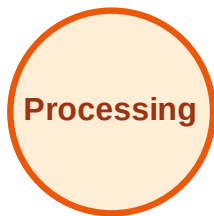
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

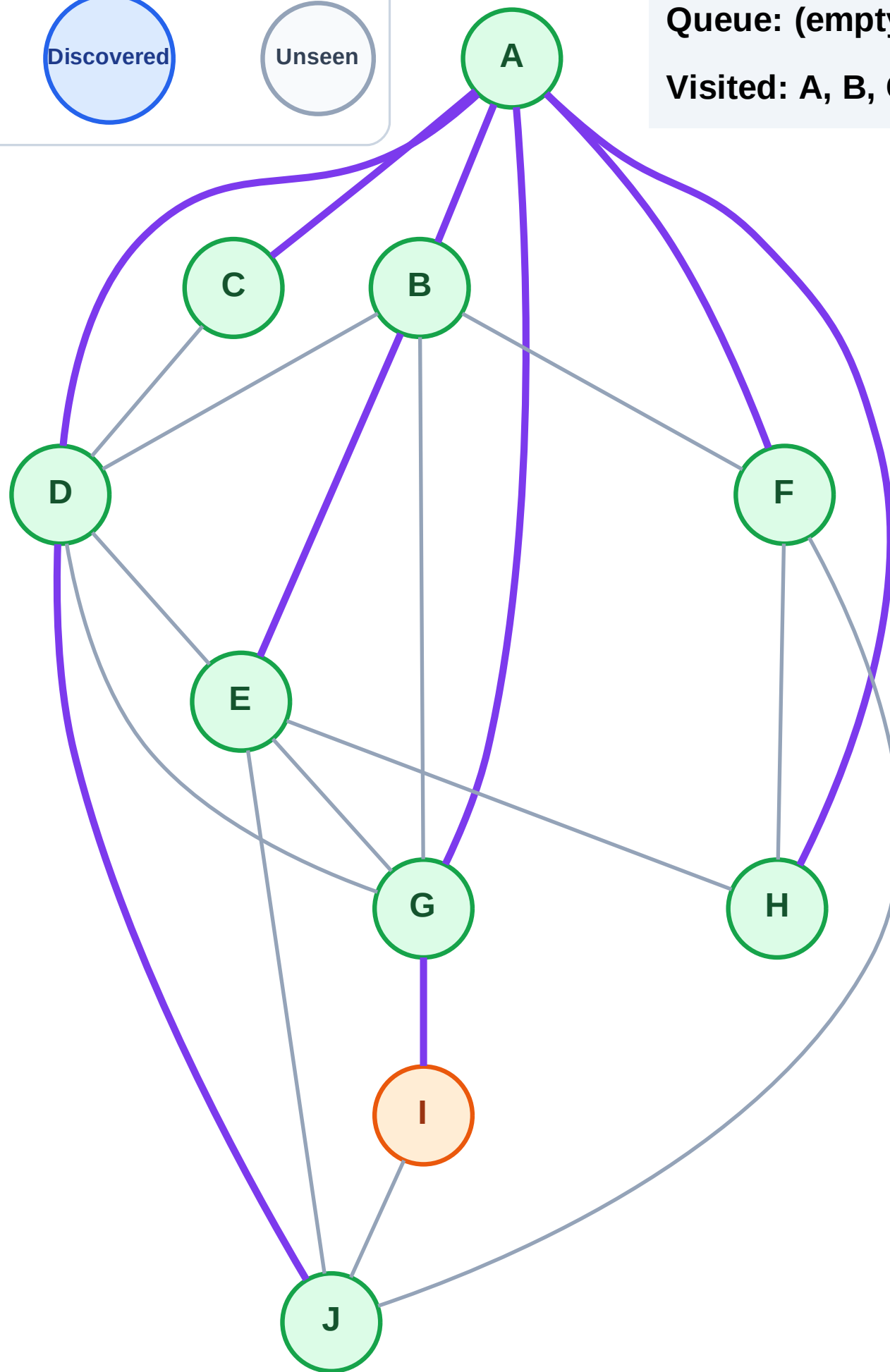
Step 20: Process node I

Legend



Queue: (empty)

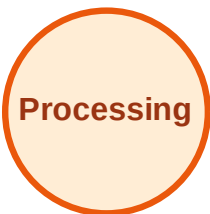
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

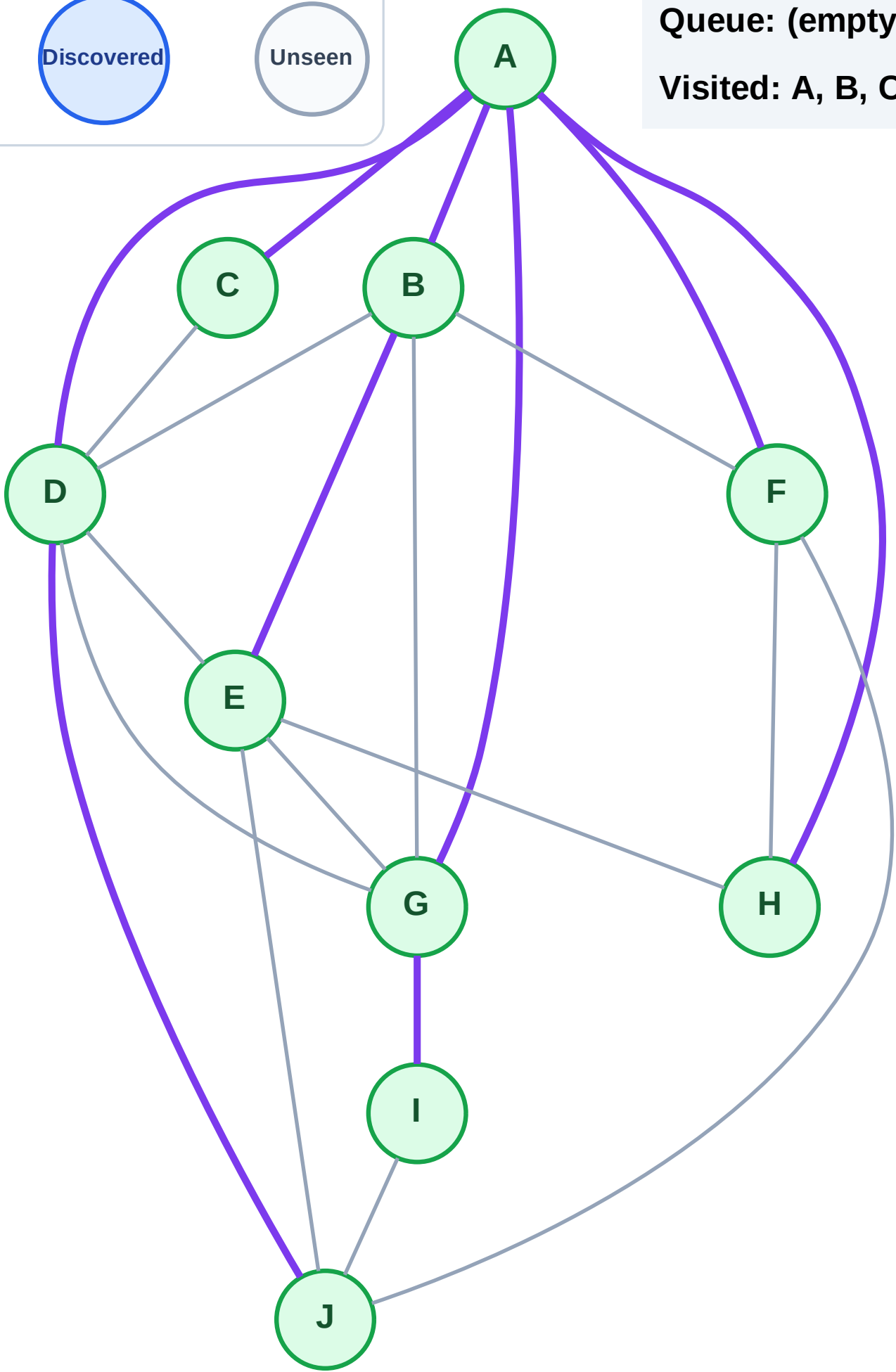
Step 21: No new neighbors from I

Legend



Queue: (empty)

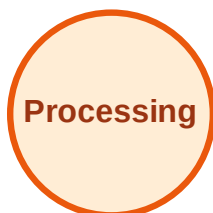
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

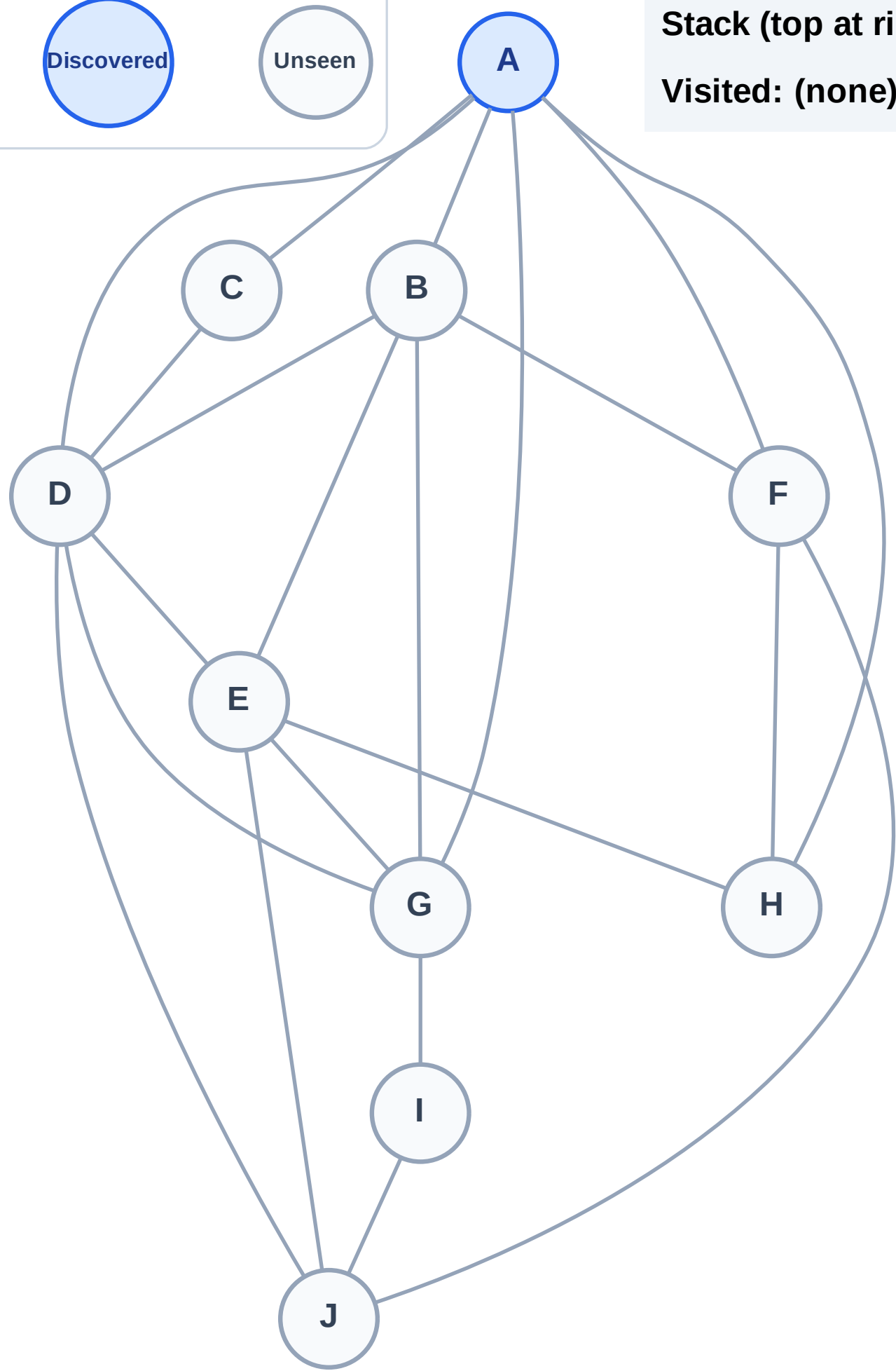
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

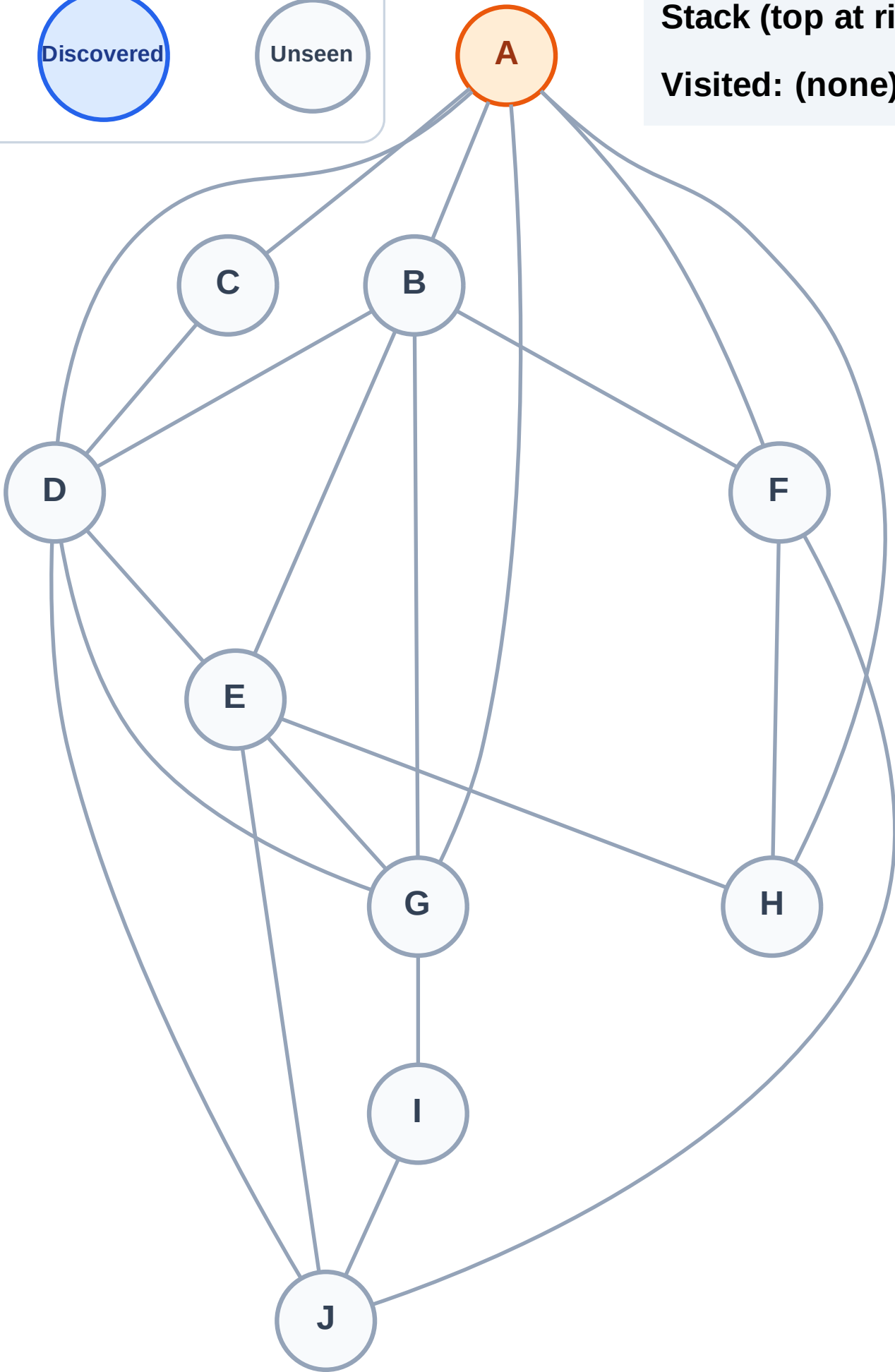
Complete

Processing

Discovered

Unseen

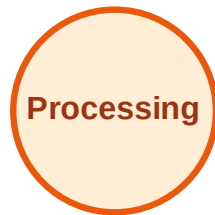
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

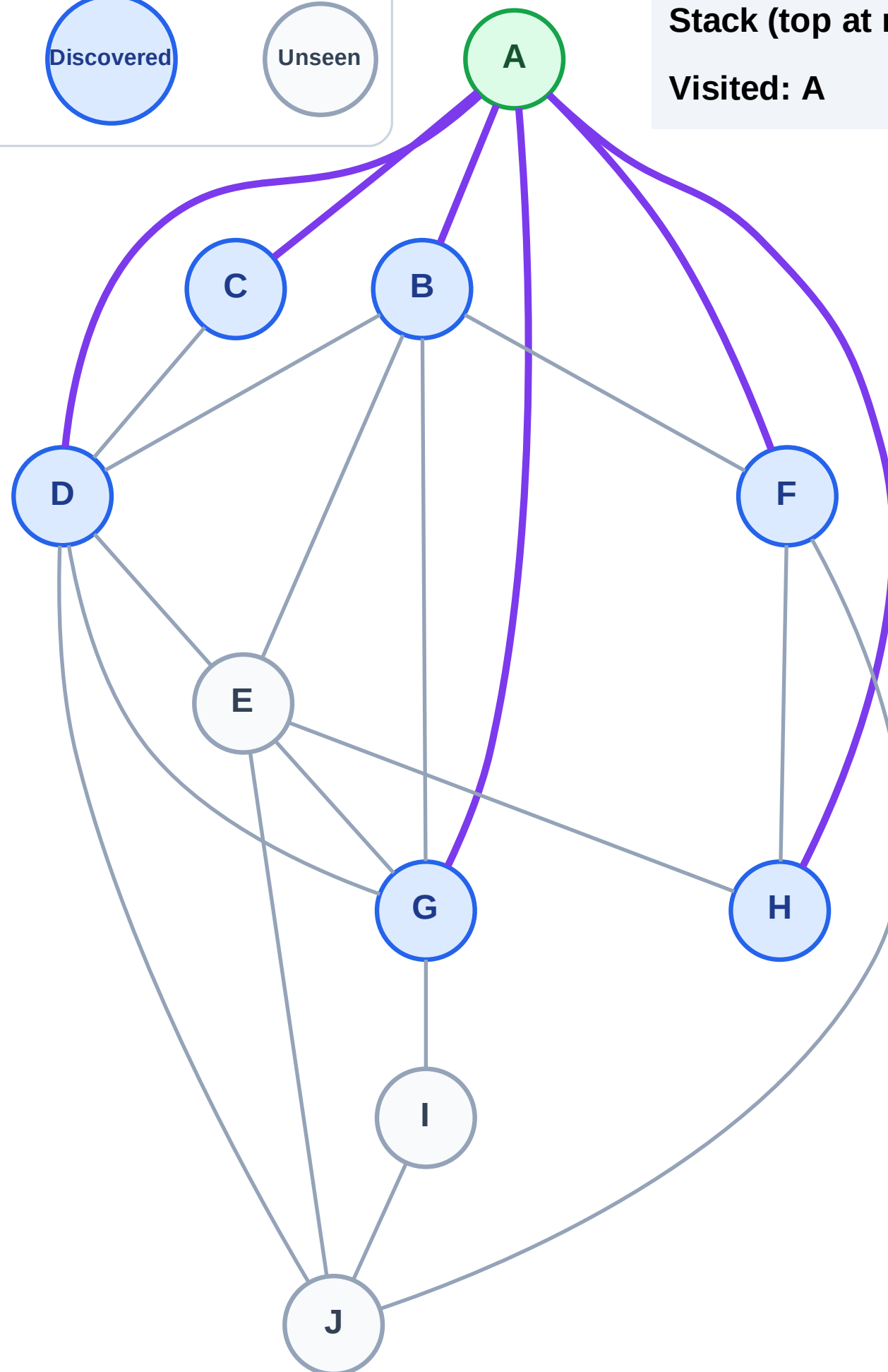
Step 3: Discover H, G, F, D, C, B from A

Legend



Stack (top at right): H | G | F | D | C | B

Visited: A



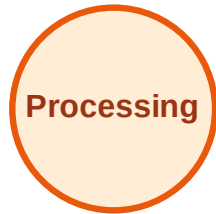
DFS Traversal

Step 4: Process node B

Legend



Complete



Processing



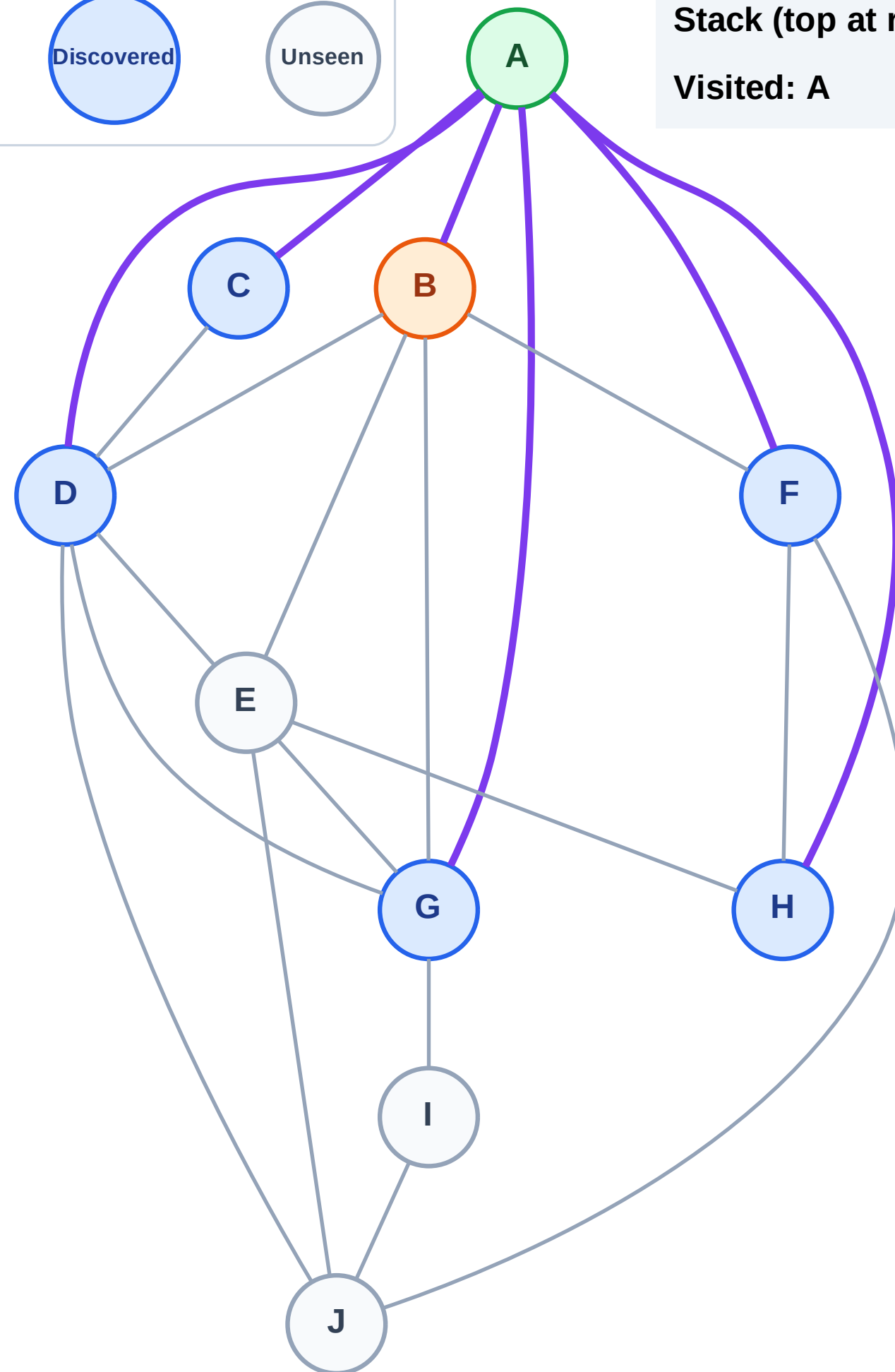
Discovered



Unseen

Stack (top at right): H | G | F | D | C

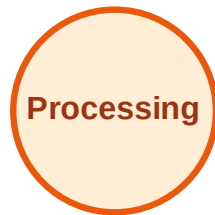
Visited: A



DFS Traversal

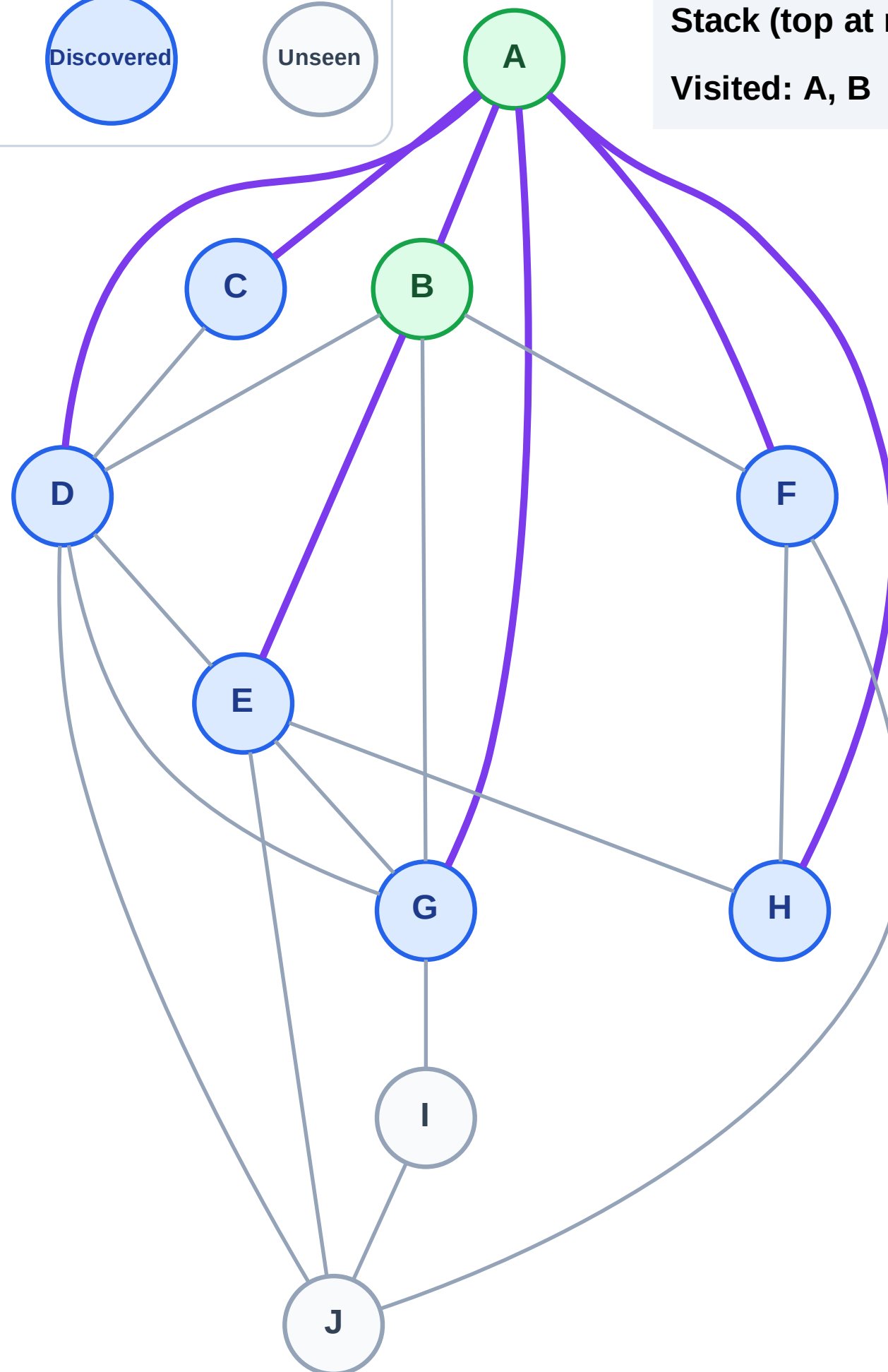
Step 5: Discover E from B

Legend



Stack (top at right): H | G | F | D | C | E

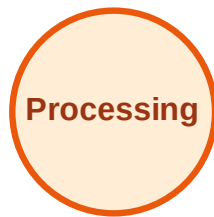
Visited: A, B



DFS Traversal

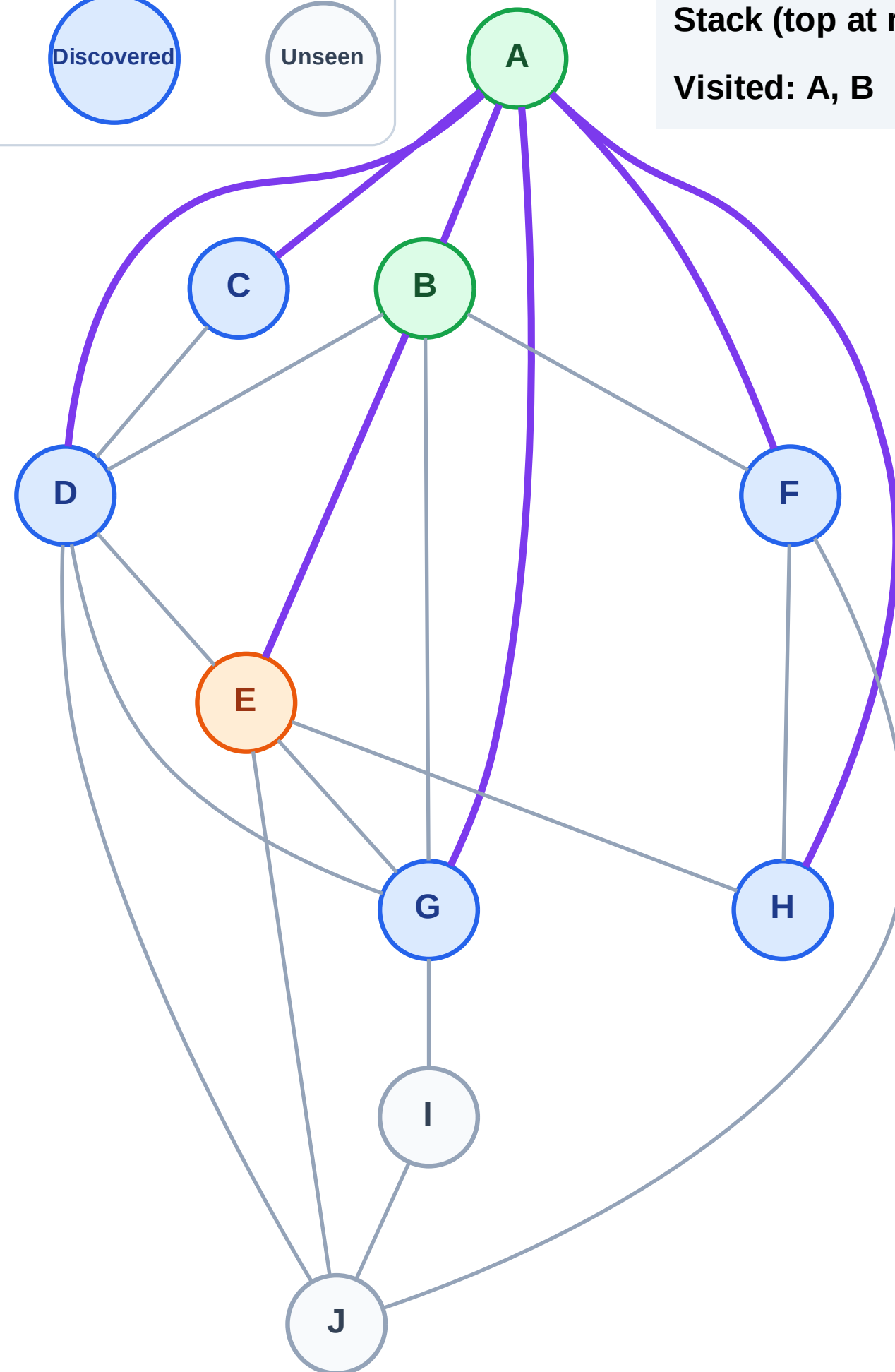
Step 6: Process node E

Legend



Stack (top at right): H | G | F | D | C

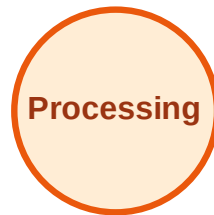
Visited: A, B



DFS Traversal

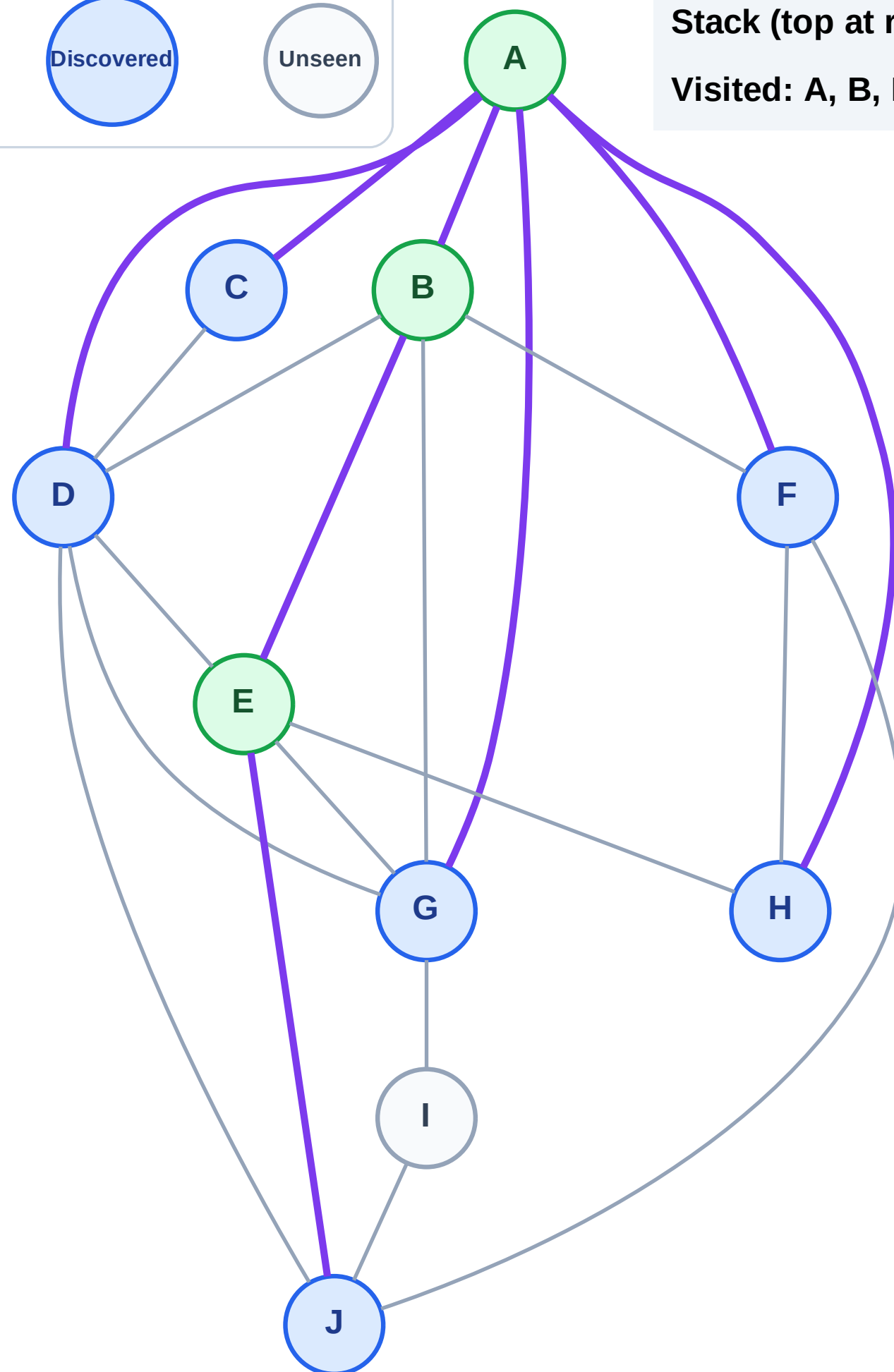
Step 7: Discover J from E

Legend



Stack (top at right): H | G | F | D | C | J

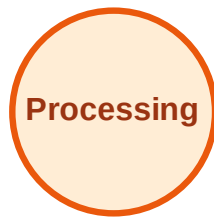
Visited: A, B, E



DFS Traversal

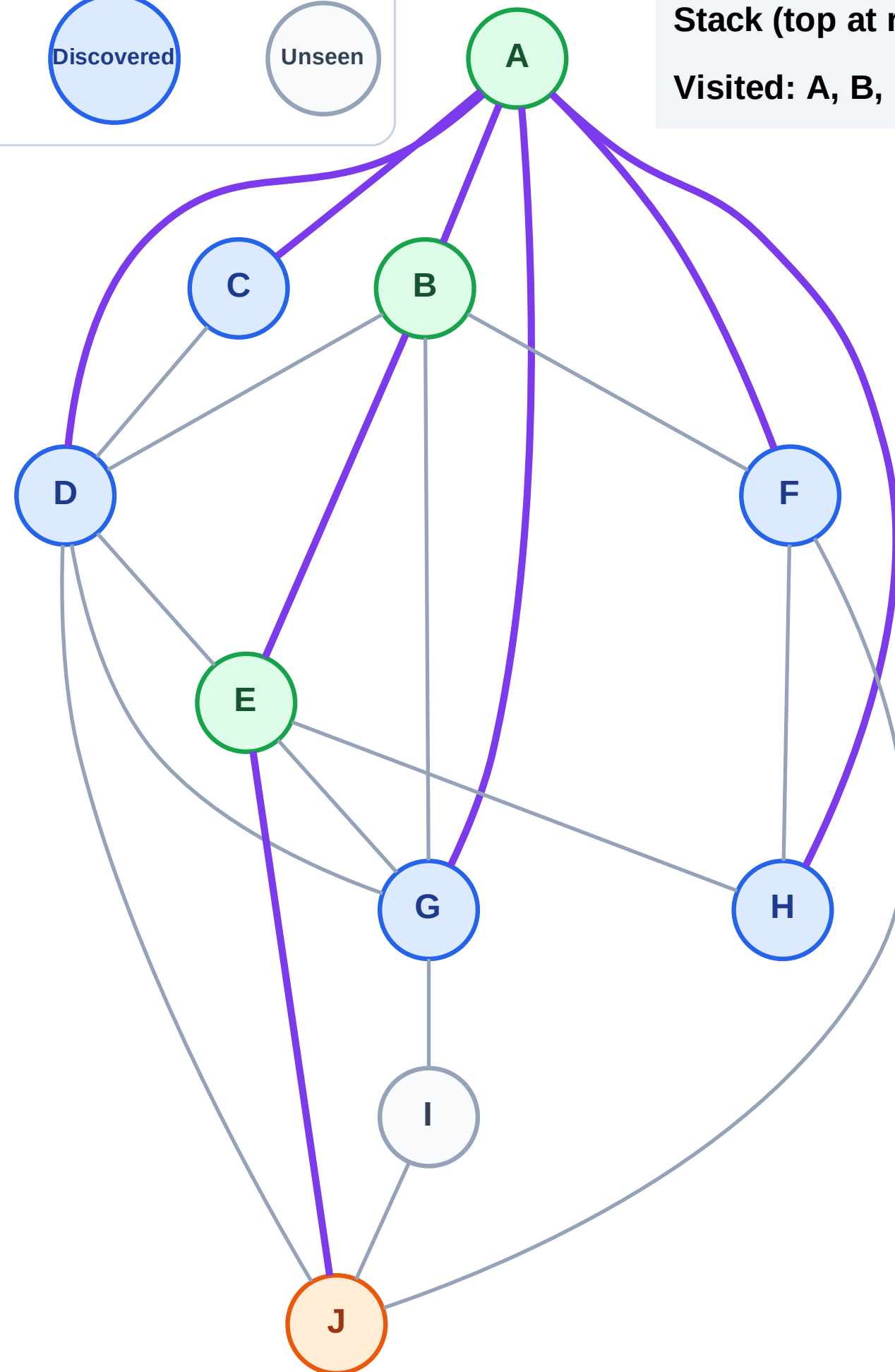
Step 8: Process node J

Legend



Stack (top at right): H | G | F | D | C

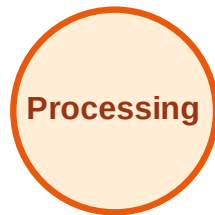
Visited: A, B, E



DFS Traversal

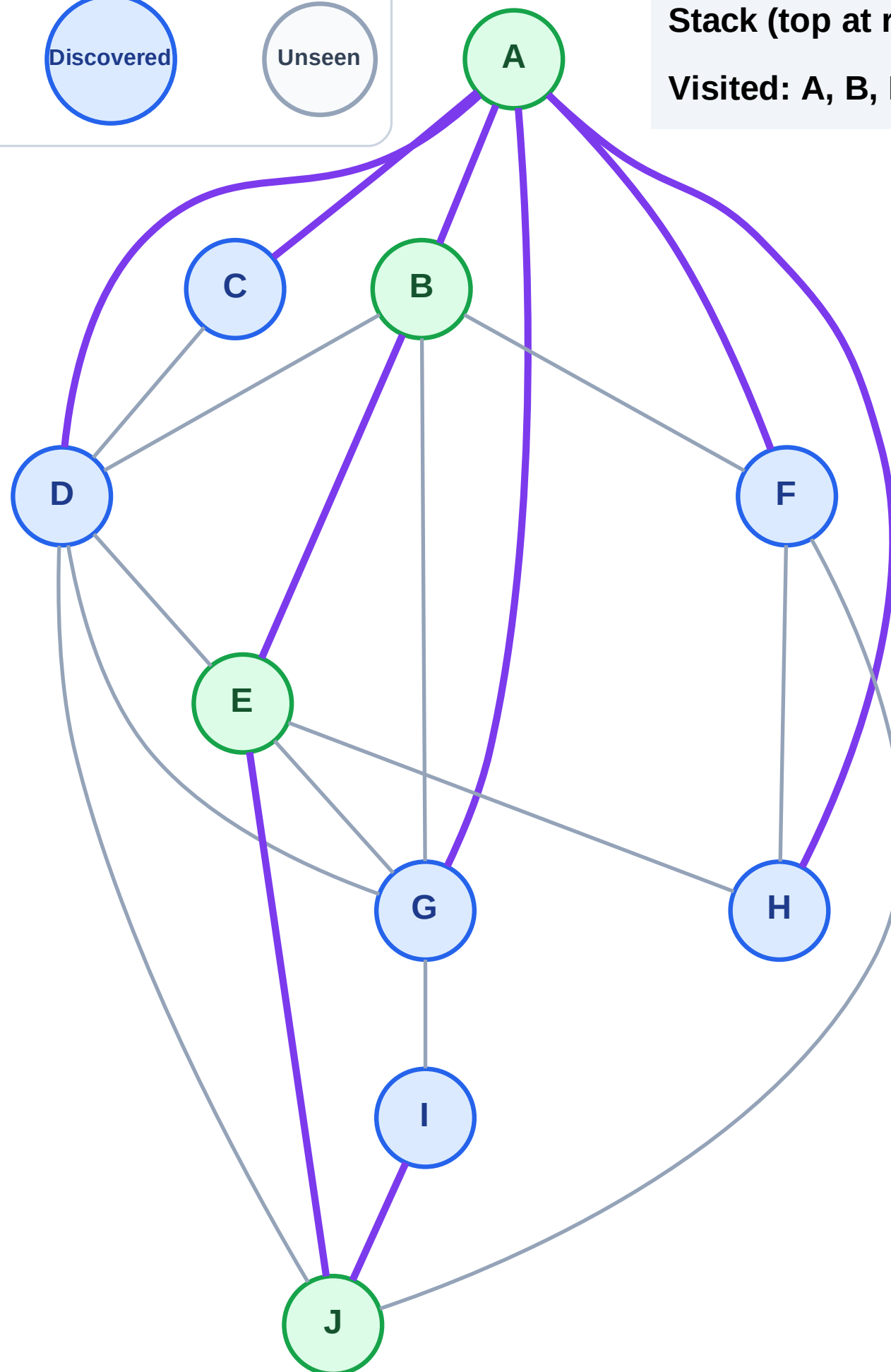
Step 9: Discover I from J

Legend



Stack (top at right): H | G | F | D | C | I

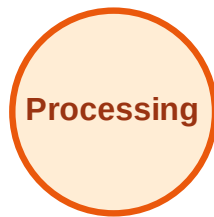
Visited: A, B, E, J



DFS Traversal

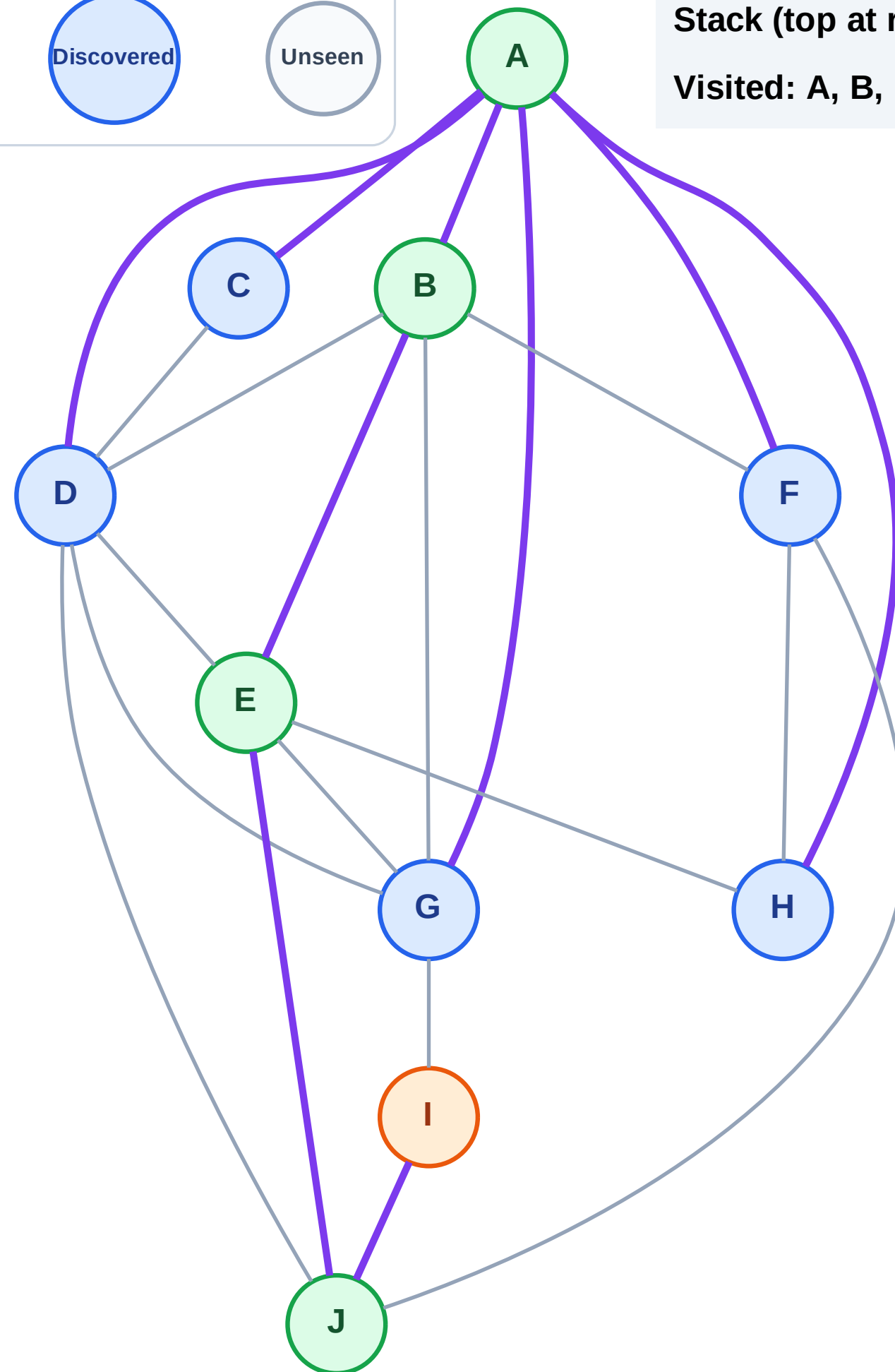
Step 10: Process node I

Legend



Stack (top at right): H | G | F | D | C

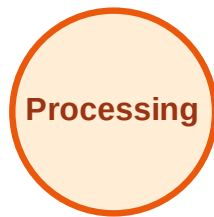
Visited: A, B, E, J



DFS Traversal

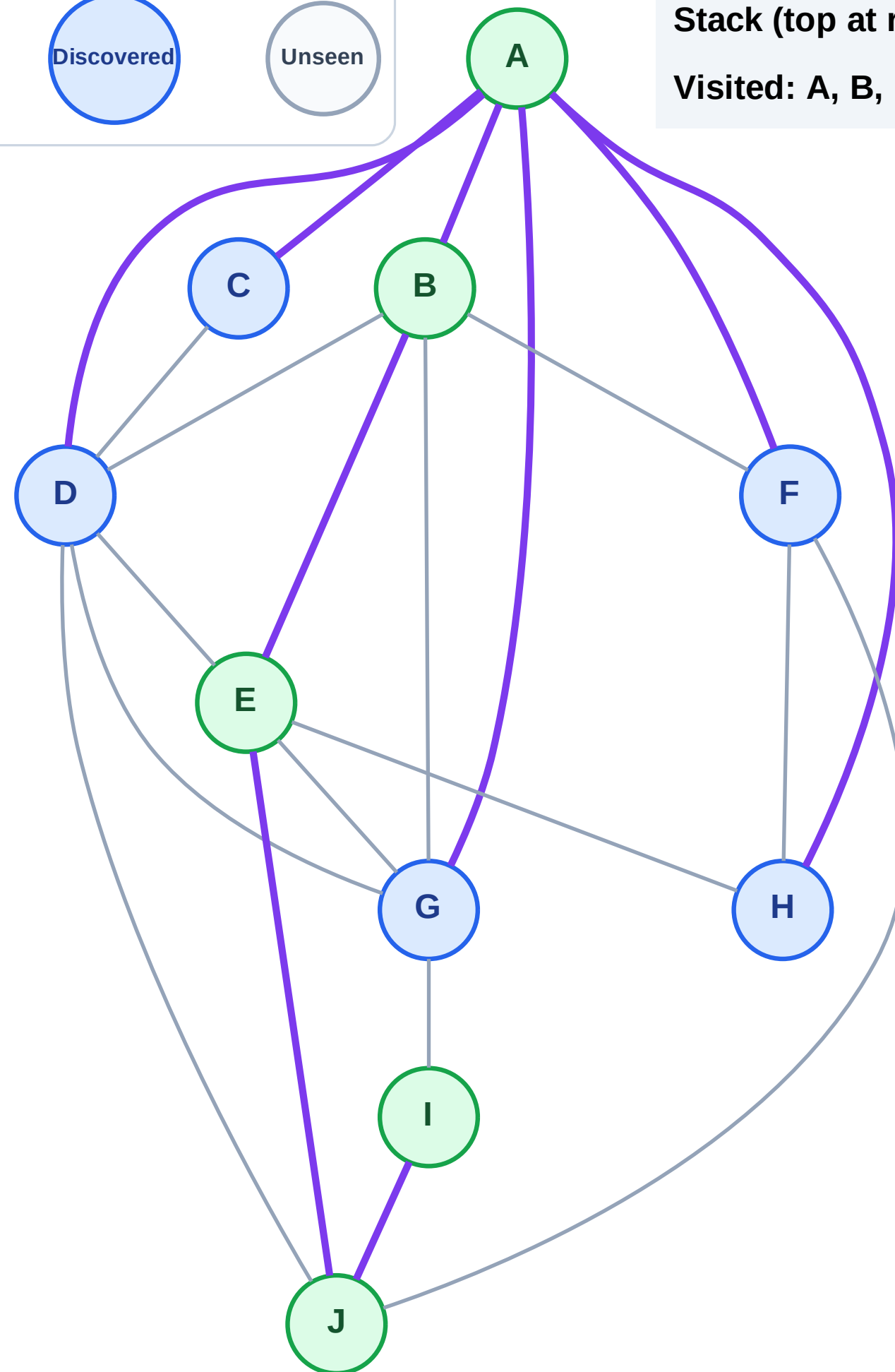
Step 11: No new neighbors from I

Legend



Stack (top at right): H | G | F | D | C

Visited: A, B, E, I, J



DFS Traversal

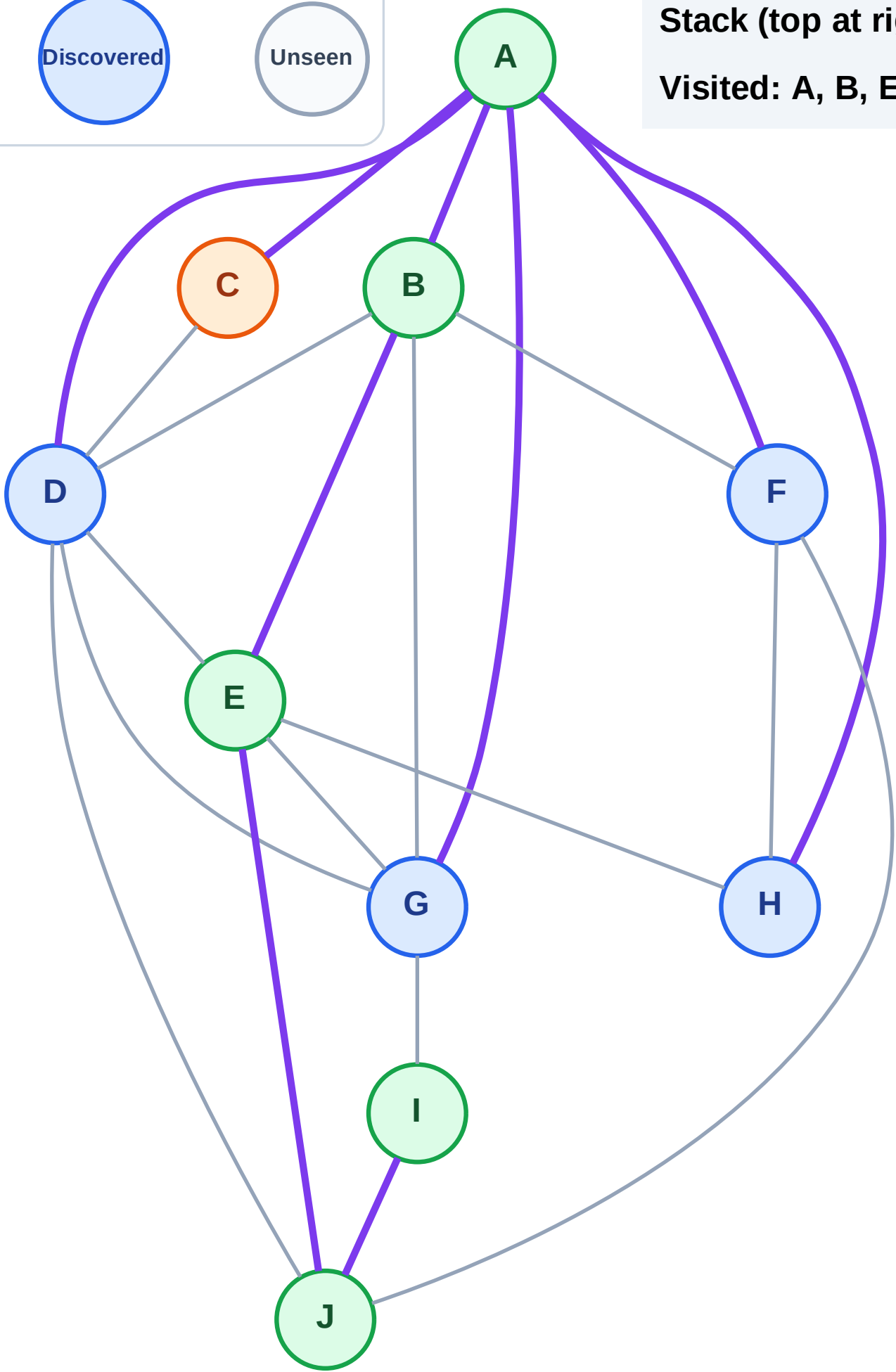
Step 12: Process node C

Legend



Stack (top at right): H | G | F | D

Visited: A, B, E, I, J



DFS Traversal

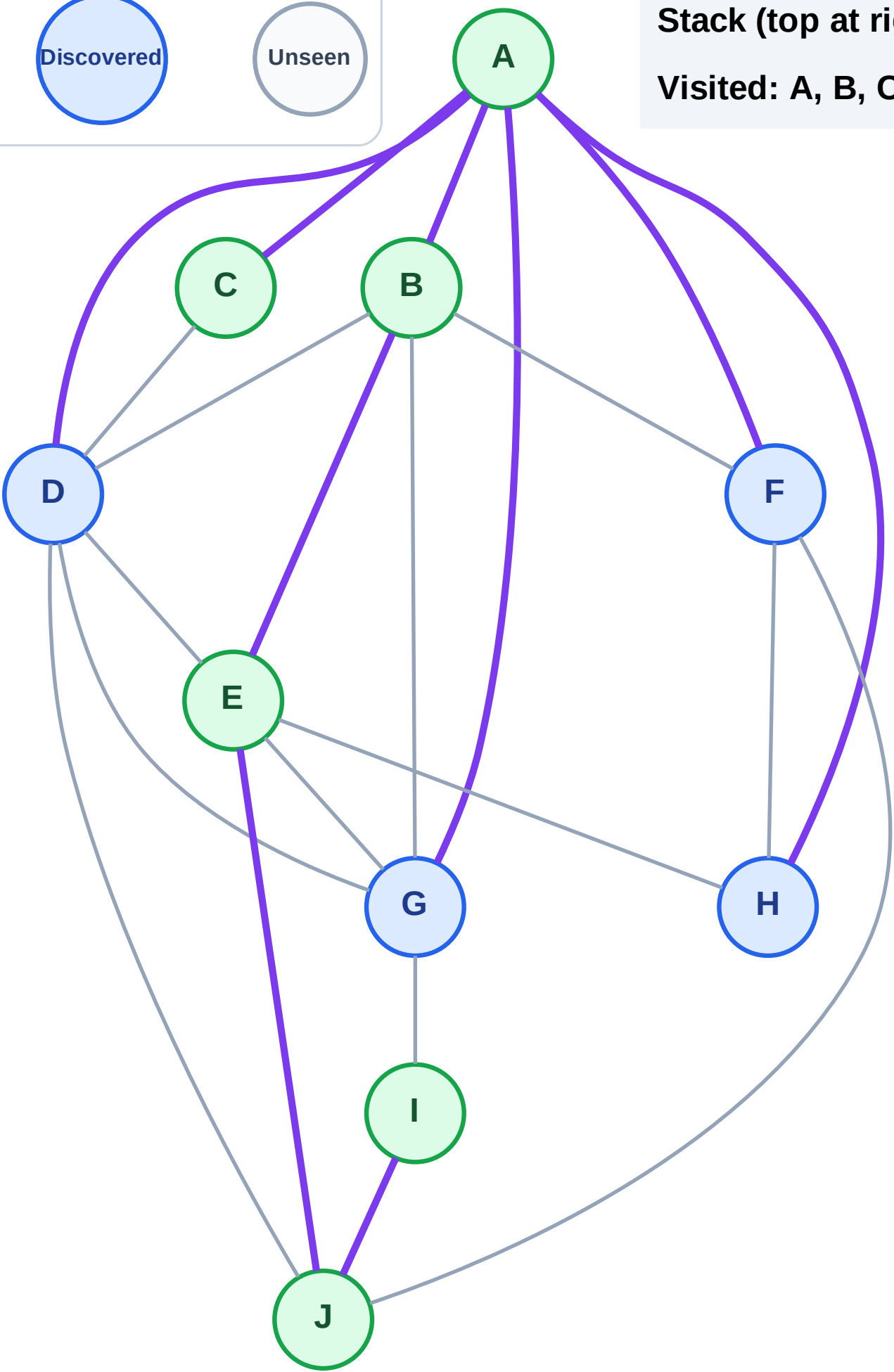
Step 13: No new neighbors from C

Legend



Stack (top at right): H | G | F | D

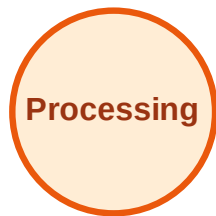
Visited: A, B, C, E, I, J



DFS Traversal

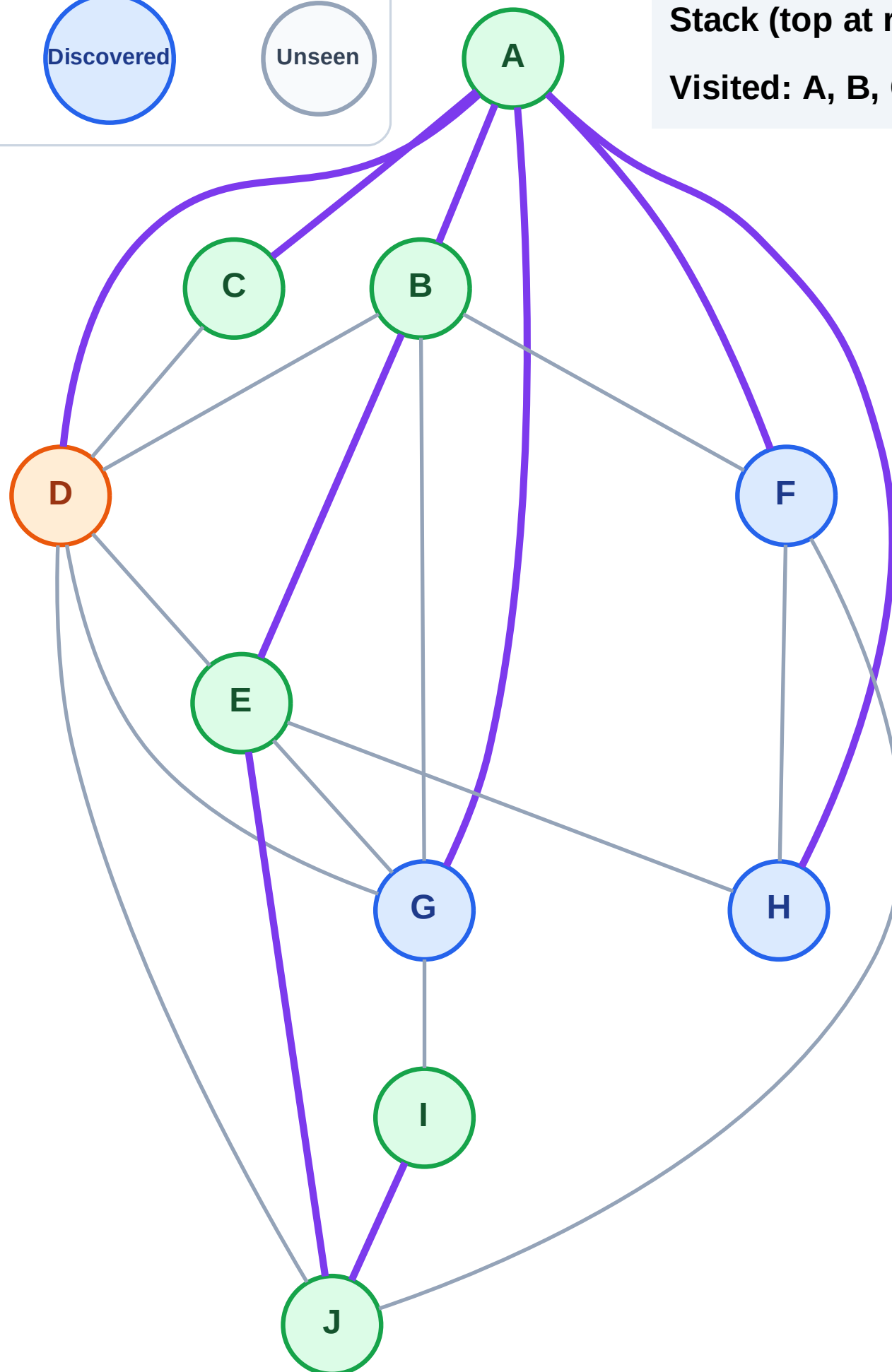
Step 14: Process node D

Legend



Stack (top at right): H | G | F

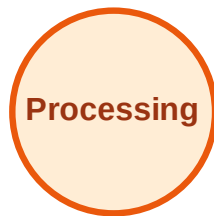
Visited: A, B, C, E, I, J



DFS Traversal

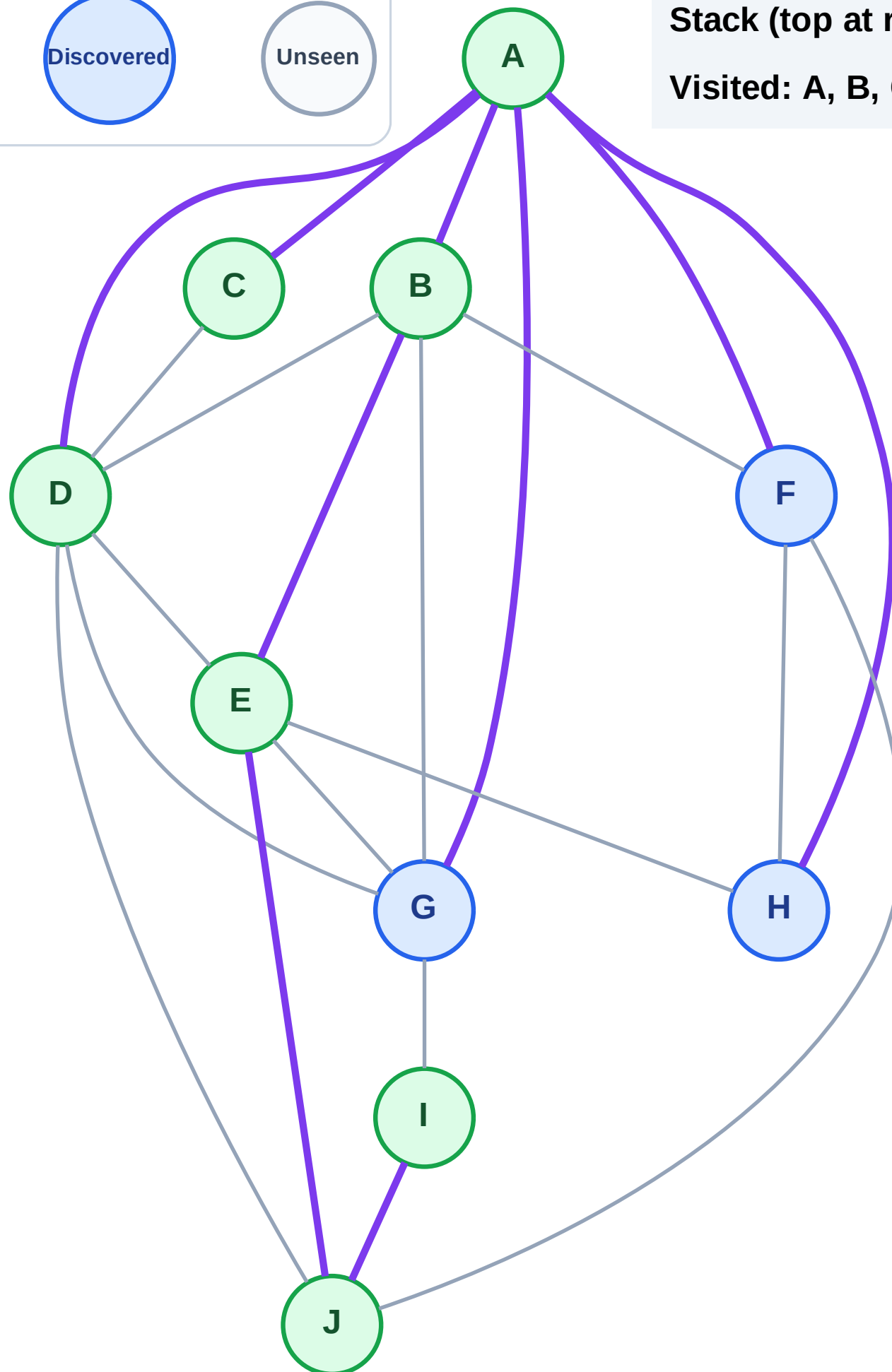
Step 15: No new neighbors from D

Legend



Stack (top at right): H | G | F

Visited: A, B, C, D, E, I, J



DFS Traversal

Step 16: Process node F

Legend

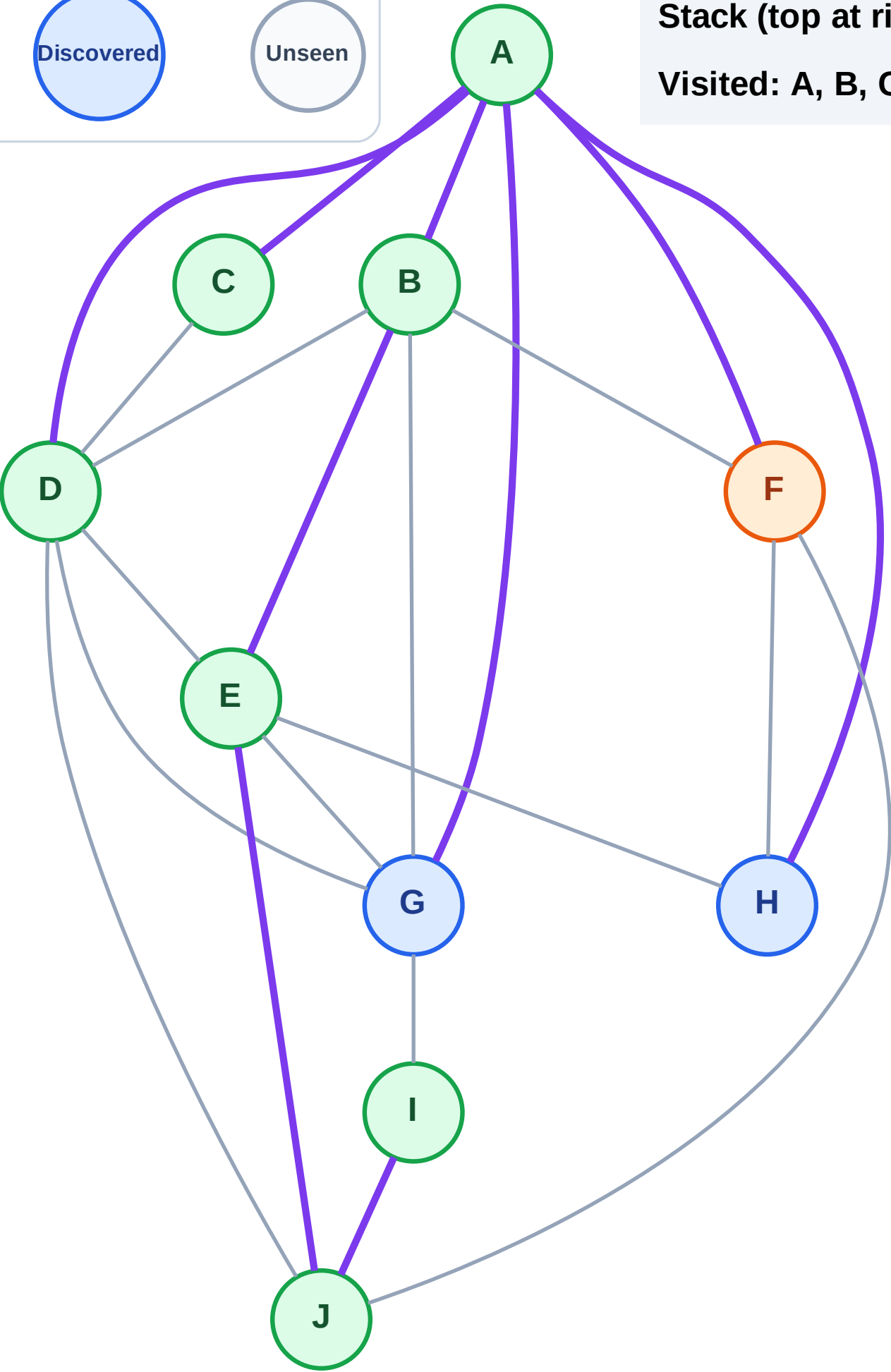
Complete

Processing

Discovered

Unseen

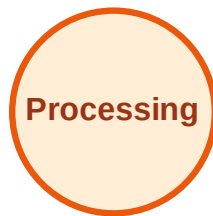
Stack (top at right): H | G
Visited: A, B, C, D, E, I, J



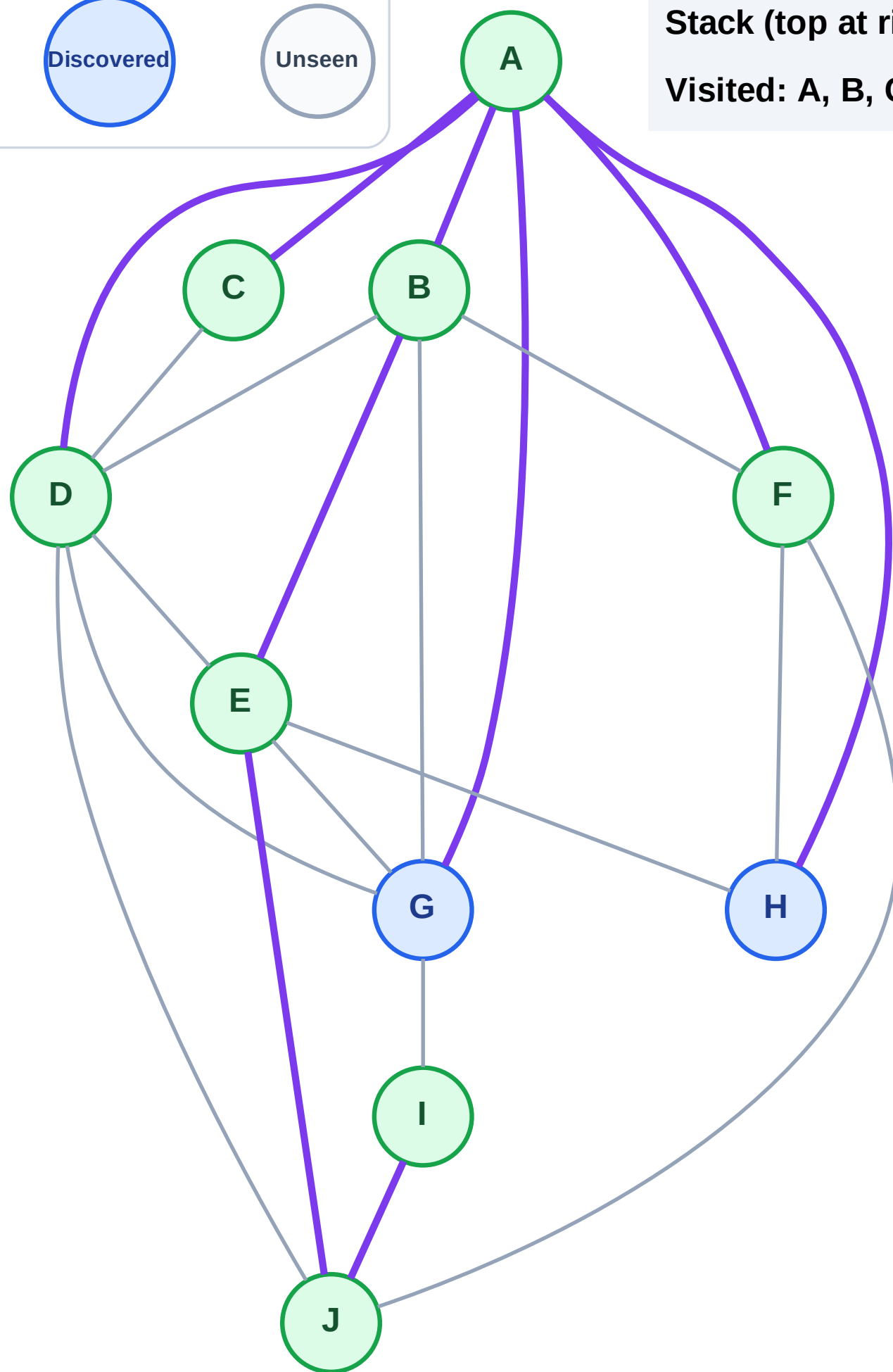
DFS Traversal

Step 17: No new neighbors from F

Legend



Stack (top at right): H | G
Visited: A, B, C, D, E, F, I, J



DFS Traversal

Step 18: Process node G

Legend

Complete

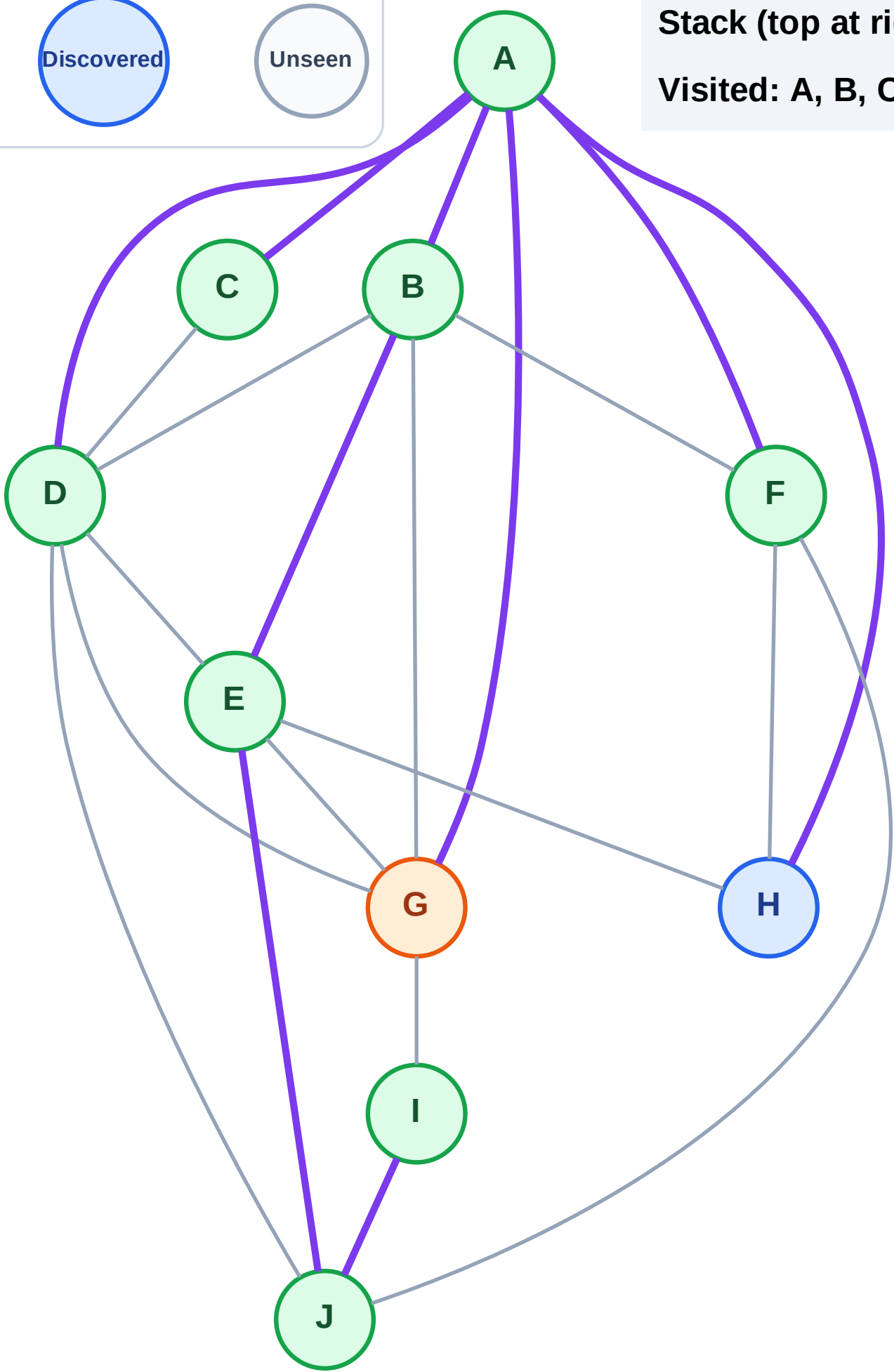
Processing

Discovered

Unseen

Stack (top at right): H

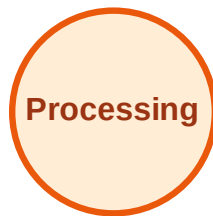
Visited: A, B, C, D, E, F, I, J



DFS Traversal

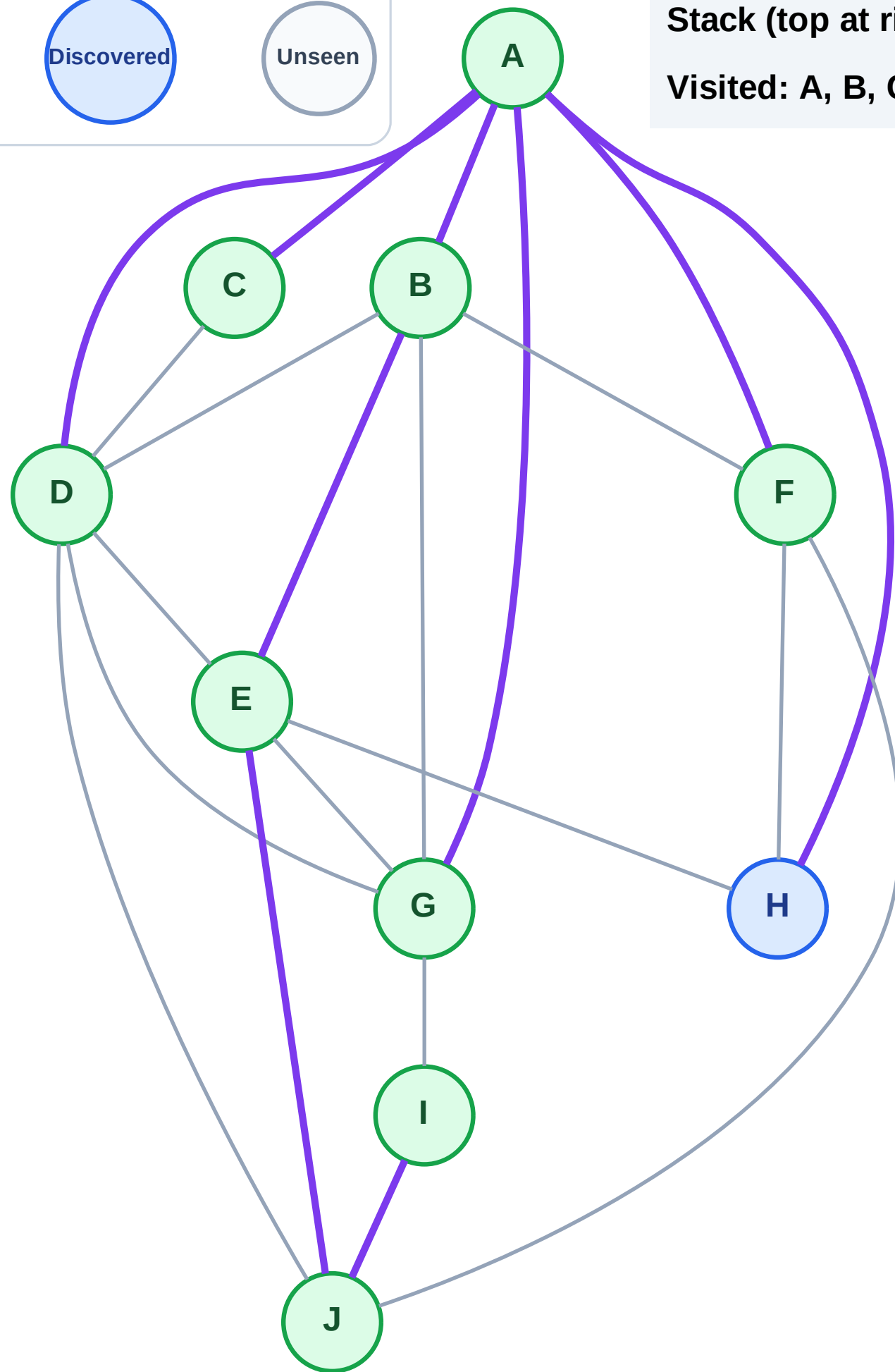
Step 19: No new neighbors from G

Legend



Stack (top at right): H

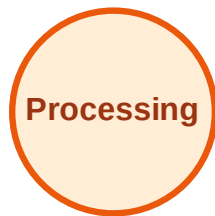
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

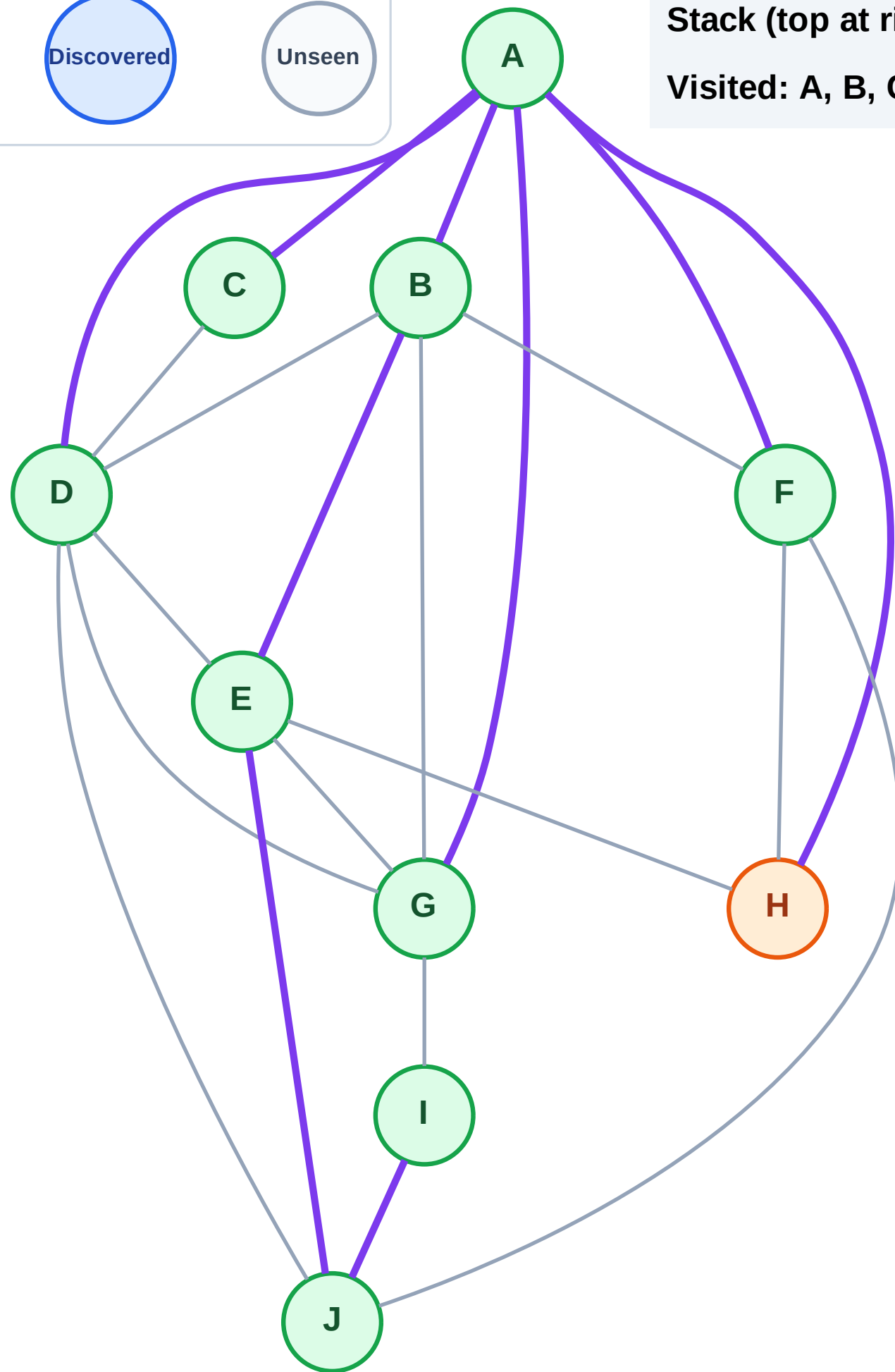
Step 20: Process node H

Legend



Stack (top at right): (empty)

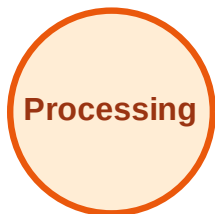
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

Step 21: No new neighbors from H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

